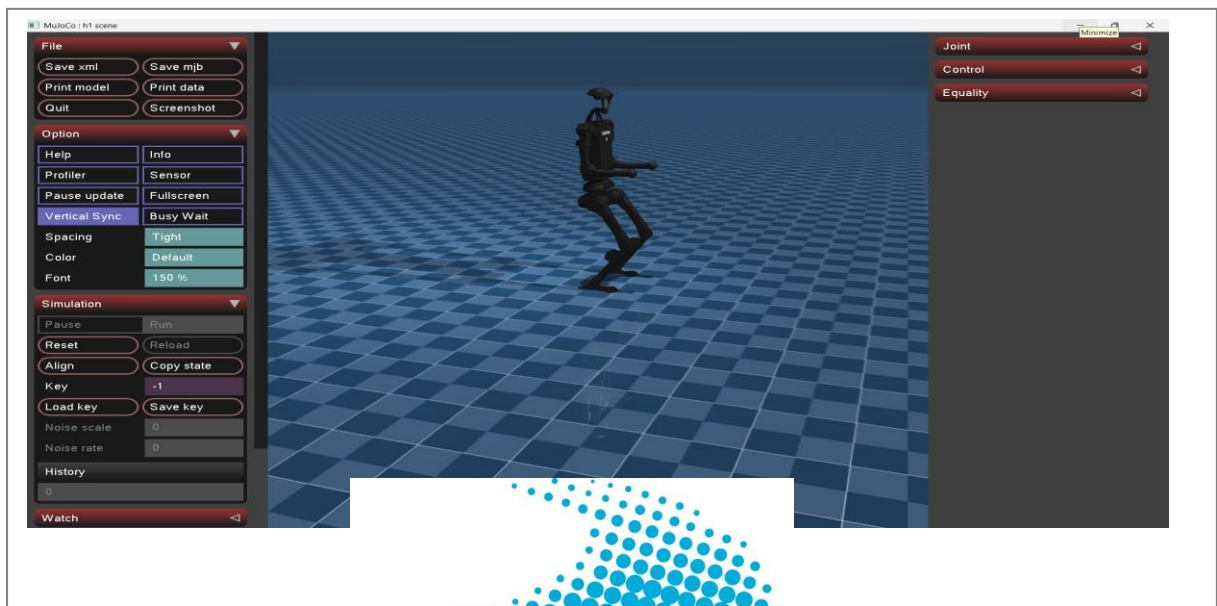


Evaluation of Whole-Body Control Architectures for Humanoid Balance in Simulation with Learning-Based Residual Compensation

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A THESIS SUBMITTED TO THE DEPARTMENT OF ENGINEERING SCIENCE
IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE DEGREE OF
MASTER OF SCIENCE WITH SPECIALIZATION IN ROBOTICS
AT UNIVERSITY WEST
2026

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Programme:	Master Programme in Robotics
Main field of study:	Automation with a specialization in industrial robotics
Credits:	30 Higher Education credits
Template:	University West, IV-Master 2.10
Publisher:	University West, Department of Engineering Science S-461 86 Trollhättan, SWEDEN Phone: + 46 520 22 30 00 Fax: + 46 520 22 32 99 Web: www.hv.se

Summary

Humanoid robots operating in human-centred environments must maintain stable balance under unexpected external disturbances. Whole-body control (WBC) provides the theoretical and computational framework for distributing corrective torques across all joints simultaneously, but the performance consequences of different WBC architectural choices under realistic disturbance conditions have not been systematically characterised in the literature. This thesis presents a simulation-based comparative evaluation of four WBC architectures — weighted quadratic programming, hierarchical quadratic programming, passivity-based control, and LQR applied to the linear inverted pendulum model — implemented for the Unitree H1 humanoid robot in MuJoCo and evaluated under a two-part disturbance protocol comprising a deterministic step-push grid and a 200-seed stochastic multi-push evaluation.

The primary finding is that weighted and hierarchical QP formulations produce statistically identical balance performance across all 200 independent trials, with a mean per-trial CoM difference of 0.50 mm, while hierarchical QP incurs 1.89 times the computational cost. The passivity-based controller produces 11% lower CoM displacement on stable trials and fewer lingering oscillation outcomes compared to both QP controllers, attributed to its energy-dissipating damping injection layer. The LQR-on-LIPM controller achieves the highest raw stability rate at low disturbance forces due to its near-zero computational overhead, but converges to the same failure floor as all other controllers above 175 N due to ankle torque saturation. A novel three-category outcome classification — stable, fell, and neither — is introduced to capture lingering oscillation outcomes absent from binary frameworks used in prior benchmarking work.

A supervised residual MLP is evaluated in progressive ablation across three training formulations. Two formulations using a LIPM prediction loss produce null results, explained theoretically by the unstable discrete-time eigenvalue of the LIPM dynamics. A third formulation incorporating five-frame temporal history and a pitch-rate behaviour cloning target produces a statistically significant improvement of 7.5 percentage points over the passivity baseline ($p < 0.001$, McNemar's exact test), generalising across all four base controllers. The improvement is largest in the 125–175 N force range, consistent with the prediction that residual learning is most effective near the base controller's feasibility boundary. These findings collectively characterise the WBC design space for static humanoid balance and establish that supervised residual learning is viable for balance augmentation when the training formulation is physically grounded, providing a practical foundation for future hardware validation and stepping controller integration.

Preface

This master's degree thesis was prepared as part of a 30 HE credits degree project within the two-year Master's Programme in Robotics and Automation at University West, Sweden, Spring 2026.

This project was conducted in parallel with a related thesis by Tharindu Dananjaya Ulpange Aluthgedara, which explores a Pinocchio-based hierarchical QP implementation with history-augmented residual learning on Ubuntu/ROS2. The literature review was developed jointly with the project partner. All controller implementations, simulation experiments, data collection, analysis, and written content in this report are the author's own independent work. Both theses are submitted as separate individual documents.

The simulation environment, robot model, and software framework were developed and configured independently by the author using publicly available open-source tools. No proprietary software, datasets, or confidential industrial materials were utilised.

The author would like to thank supervisor Xiaoxiao Zhang for guidance and feedback throughout the project, including project planning, experimental design, and thesis writing, and examiner Prabhu Kettavarampalayam Ramanathan for feedback on the project plan.

Affirmation

This master degree report, *Evaluation of Whole-Body Control Architectures for Humanoid Balance in Simulation with Learning-Based Residual Compensation*, was written as part of the master degree work needed to obtain a Master of Science with specialization in Robotics degree at University West. All material in this report, that is not my own, is clearly identified and used in an appropriate and correct way. The main part of the work included in this degree project has not previously been published or used for obtaining another degree.

12-05-2026

Signature by the author

Date

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Symbols and glossary

α	Stress weighting factor used in the MLP v3 behaviour cloning target. Computed as $\text{clip}(\dot{\theta}_{\text{pitch}} / 1.0, 0, 1)$, it scales the training target torque proportionally to the current torso pitch rate, ensuring the target correction is non-zero only during active disturbance events and zero during steady-state standing.
b (Damping gain)	The positive scalar gain applied to the velocity-proportional damping injection term in the passivity-based controller, $\tau_{\text{damping}} = -b \cdot \dot{q}$. Ensures the controller dissipates kinetic energy at every timestep, satisfying the passivity condition. Set to 20.0 Nm·s/rad in this thesis.
$e_{\text{pitch}}, e_{\text{roll}}$	Balance error signals in the sagittal and frontal planes respectively, used as inputs to the QP balance objective. Computed as a proportional-derivative function of the torso pitch and roll angles relative to their reference values, following the ankle strategy formulation of Stephens (2007).
e_{posture}	Joint angle deviation vector between the current joint positions and the nominal standing configuration. Used as the posture task error in the WBC quadratic program. Minimising this term keeps the robot close to its reference standing posture.
g (Gravitational acceleration)	The standard gravitational acceleration constant, 9.81 m/s ² . Used in the LIPM dynamics equation $\ddot{x}_{\text{CoM}} = (g / z_c)(x_{\text{CoM}} - x_{\text{CoP}})$ and in the LQR ankle torque relationship $\tau_{\text{ankle}} = m \cdot g \cdot \text{CoP}_{\text{cmd}}$.
H (QP cost matrix)	The positive definite matrix defining the quadratic cost in the WBC optimisation problem. Encodes the weighted task tracking objectives as a single matrix expression $(1/2)\tau^T H \tau + c^T \tau$, where c is the linear cost vector.
m (Robot total mass)	The total mass of the Unitree H1 humanoid, 47 kg. Used in the LQR ankle torque computation $\tau_{\text{ankle}} = m \cdot g \cdot \text{CoP}_{\text{cmd}}$ and in the LIPM dynamics where the robot is approximated as a single point mass.
N (Null-space projector)	A matrix that projects lower-priority task objectives into the null space of a higher-priority task Jacobian, ensuring the lower-priority task does not

affect the higher-priority solution. Used in the Operational Space Formulation to achieve hierarchical task execution.

$\mathbf{p}, \mathbf{L}$ (Linear and angular momentum)	The linear momentum vector $\mathbf{p}$ and angular momentum vector $\mathbf{L}$ of the full robot computed at the centre of mass, collectively termed centroidal momentum. Regulating these quantities as a high-priority balance task is the approach demonstrated by Herzog et al. (2016).
τ_{base}	The torque vector produced by the base model-based controller at each control step, before the residual MLP correction is applied. Forms one input to the addition block $\tau_{\text{total}} = \tau_{\text{base}} + \delta\tau$ and is also provided as a contextual feature in each frame of the MLP input.
τ_{gravity}	The gravity and Coriolis compensation torque vector, obtained from MuJoCo's <code>qfric_bias</code> field at each control step. Represents the torques required to hold the robot stationary against gravity in its current configuration. Included in all four controllers evaluated in this thesis.
τ_{total}	The final joint torque vector sent to the MuJoCo actuators at each control step, defined as $\tau_{\text{total}} = \tau_{\text{base}} + \delta\tau$ when the residual MLP is active, or $\tau_{\text{total}} = \tau_{\text{WBC}}$ when operating with the base controller alone.
V (Lyapunov function)	An energy-like scalar function of the system state used to prove stability in the passivity-based controller. The passivity condition requires $\dot{V} \leq -\alpha \ \mathbf{x}\ ^2$, guaranteeing that the stored energy decreases over time and the system converges toward equilibrium.
W (Task weight matrix)	A diagonal matrix whose entries define the relative importance of each task in the WQP cost function. In the controllers evaluated in this thesis, the balance task weight is set to 2000 and the posture task weight to 1.0, producing a 2000:1 ratio that makes balance effectively dominant.
$x_{\text{CoM}}, \dot{x}_{\text{CoM}}$	Horizontal CoM position and velocity in the sagittal plane, forming two components of the four-dimensional LIPM state vector used by the LQR controller. The full state is $[x_{\text{CoM}}, \dot{x}_{\text{CoM}}, y_{\text{CoM}}, \dot{y}_{\text{CoM}}]^T$ covering both sagittal and frontal planes.
x_{CoP}	Commanded centre of pressure position in the sagittal plane, the control input to the LIPM system. Computed by the LQR as $x_{\text{CoP}} = -K \cdot [x_{\text{CoM}}, \dot{x}_{\text{CoM}}, y_{\text{CoM}}, \dot{y}_{\text{CoM}}]^T$ and converted to ankle torque via $\tau_{\text{ankle}} = m \cdot g \cdot x_{\text{CoP}}$.
z_c (CoM height)	The nominal height of the centre of mass above the ground in the LIPM model, set to 0.9504 m corresponding to the Unitree H1 home keyframe configuration. Determines the LIPM natural frequency $\omega_0 = \sqrt{g / z_c}$ and the LQR gain matrix computed at initialisation.
BC (Behaviour Cloning)	A supervised learning method in which a neural network is trained to imitate a target behaviour by minimising the difference between its output and a set of labelled training examples. Used here as the training paradigm

for MLP v3, where the target is derived from the ankle strategy of Stephens (2007) rather than from expert demonstrations.

Capture Point	The unique point on the ground to which the robot must step to bring its CoM velocity to zero without further control input, derived analytically from the LIPM dynamics by Pratt et al. (2006). Defines the boundary between ankle-strategy recovery, where no step is needed, and step-strategy recovery, where ankle torques alone are insufficient to prevent falling.
CoM (Centre of Mass)	The weighted average position of all rigid body segments of the robot, computed from body masses and positions. Deviation of the CoM from its reference position above the support polygon is the primary measure of balance disturbance used throughout this thesis.
CoP (Centre of Pressure)	The point on the ground support surface where the net ground reaction force effectively acts. In the LIPM framework, commanding the CoP position determines the CoM acceleration. The ankle joints command the CoP by generating moments about the foot contact point.
$\delta\tau$ (Residual torque correction)	The additive torque correction produced by the residual MLP at each control step, bounded to ± 20 Nm per joint. Added to the base controller output τ_{WBC} to form the final torque command $\tau_{\text{total}} = \tau_{\text{WBC}} + \delta\tau$. Applied only to the six balance-critical joints; all other joints receive the base controller torque unchanged.
HQP (Hierarchical Quadratic Programming)	A whole-body control solver architecture in which multiple tasks are arranged in a strict lexicographic priority order. Each lower-priority task is solved only within the solution space of all higher-priority tasks, guaranteeing that a lower-priority objective can never degrade a higher-priority one. Contrasts with WQP, where all tasks compete in a single weighted cost function.
J (Task Jacobian)	A matrix that maps joint velocities to task-space velocities in the Operational Space Formulation. Its transpose J^T maps task-space force commands to joint torques, enabling task-space objectives to be expressed as joint torque requirements.
K (LQR gain matrix)	The optimal feedback gain matrix computed by solving the discrete algebraic Riccati equation for the LIPM system. Maps the four-dimensional CoM state error — horizontal position and velocity in both sagittal and frontal planes — to a commanded centre of pressure correction.
LIPM (Linear Inverted Pendulum Model)	A simplified dynamic model that approximates the full humanoid as a point mass constrained to move at constant height above the ground. The resulting CoM dynamics are linear, enabling analytical solutions and real-time control design. The constant-height assumption is the primary source of model error for the LQR-on-LIPM controller at large disturbance magnitudes.

LQR (Linear Quadratic Regulator)	An optimal control method that computes a feedback gain matrix minimising a quadratic cost function over state deviation and control effort. Applied here to the discretised LIPM dynamics to compute ankle torque commands from CoM state error, following the formulation of Stephens (2007).
MLP (Multi-Layer Perceptron)	A feedforward neural network consisting of an input layer, one or more hidden layers with nonlinear activation functions, and an output layer. Used here as the residual correction module mapping a 260-dimensional history-augmented state to a six-dimensional torque correction bounded to ± 20 Nm.
Neither (outcome category)	A trial outcome classification introduced in this thesis in which the robot neither falls nor fully recovers to the reference posture within the 10-second evaluation window. Represents a locally stable but displaced quasi-equilibrium not captured by the binary stable/fell classification used in prior benchmarking work.
OSF (Operational Space Formulation)	A control framework introduced by Khatib (1987) that specifies desired robot behaviour as forces or accelerations in task space and maps them to joint torques via the robot's mass matrix and task Jacobians. The theoretical foundation of all four WBC architectures evaluated in this thesis.
$q, \dot{q}, \ddot{q}$	Joint position vector, joint velocity vector, and joint acceleration vector respectively, each of dimension equal to the number of actuated degrees of freedom. For the Unitree H1 evaluated in this thesis, each vector has 19 elements corresponding to the 19 actuated joints.
QP (Quadratic Programming)	A class of mathematical optimisation problem with a quadratic objective function and linear constraints. Used in WBC to compute optimal joint torques that minimise task tracking errors subject to torque limits and contact constraints. Solved at each 100 Hz control step in the WQP and HQP controllers evaluated in this thesis.
τ (Joint torque vector)	The primary decision variable in the WBC optimisation, representing the vector of torques commanded to all actuated joints at each control step. The WBC computes τ by solving a quadratic program that minimises task tracking errors subject to torque limit constraints.
τ_{rms} (Torque RMS)	The root-mean-square of all joint torques computed over the duration of a trial. Measures the average control effort expended by the controller and is used as an energy efficiency metric in the comparative evaluation of this thesis.
WBC (Whole-Body Control)	An optimisation-based control framework that computes joint torques for all degrees of freedom of a humanoid robot simultaneously, satisfying multiple task objectives — such as balance and posture regulation — subject to physical constraints including joint torque limits and contact force feasibility.

WQP (Weighted Quadratic Program)	A whole-body control formulation in which all task objectives are combined into a single quadratic cost function with numerical weighting coefficients and solved simultaneously. The weight ratio between tasks determines relative priority, but lower-priority tasks can always partially degrade higher-priority ones when objectives conflict — the fundamental limitation that HQP is designed to address.
ZMP (Zero Moment Point)	The point on the ground support surface at which the resultant ground reaction force produces zero overturning moment. Equivalent to the CoP for flat-footed contact. Used as a stability criterion in classical gait planning: the robot is statically stable if and only if the ZMP lies within the support polygon.
ω_0 (LIPM nat- ural fre- quency)	Defined as $\sqrt{g / z_c}$, where g is gravitational acceleration and z_c is the CoM height. Determines the time constant of the LIPM dynamics and the capture point distance from the CoM for a given CoM velocity.

1 Introduction

1.1 Background and Motivation

Humanoid robots are increasingly considered for deployment in industrial and domestic environments where stable, disturbance-resilient balance is a fundamental operational requirement. In manufacturing facilities, logistics centres, and healthcare settings, a stationary humanoid must withstand physical contact arising from collisions with moving equipment, accidental pushes from personnel, or sudden payload changes. The failure to maintain balance under such conditions not only renders the robot operationally unreliable but introduces significant safety risks in proximity to people. This observation is not merely theoretical: balance under external disturbance has been identified in recent comprehensive surveys of humanoid control as the foundational requirement from which all higher-level task execution depends [1].

The control component responsible for this capability is the Whole-Body Controller (WBC). At each control timestep, the WBC distributes torque commands across all actuated joints simultaneously, satisfying multiple competing objectives — maintaining upright posture, regulating the centre of mass (CoM) above the support polygon, and respecting joint torque limits and contact constraints — without violating the robot's full-body dynamics. The theoretical foundation for this class of controller was established by Khatib [2] through the Operational Space Formulation, which maps task-space objectives to joint torques via the robot's mass matrix and Jacobians, enabling unified motion and force control across all degrees of freedom. Extensions of this framework produced a range of architectures that differ in how competing objectives are prioritised and how the resulting optimisation is solved. Escande and Mansard [3] formalised the Hierarchical Quadratic Programming (HQP) framework, in which tasks are arranged in a strict lexicographic priority order so that higher-priority objectives are never compromised by lower-priority ones. Herzog et al. [4] demonstrated hierarchical inverse dynamics control on a real torque-controlled humanoid, showing that centroidal momentum regulation can serve as a high-priority balance task while posture objectives are resolved in the remaining null space. Henze et al. [5] introduced a passivity-based formulation as an alternative, deriving torques analytically from energy-dissipating Lyapunov functions rather than solving an optimisation problem, and demonstrating compliant, passive behaviour in multi-contact balancing scenarios on the DLR TORO robot.

Despite the maturity of WBC as a field, systematic comparative evaluation of these architectural variants under identical, controlled disturbance conditions remains limited. Most published evaluations test a single architecture on a specific hardware platform under disturbance protocols that are not standardised across studies, making cross-architecture comparison unreliable. Sovukluk et al. [6] provide one of the few direct experimental comparisons between WBC paradigms — specifically between inverse-dynamics and passivity-based formulations — on a shared hardware platform, finding that the passivity formulation exhibits superior robustness to modelling errors while inverse-dynamics achieves more precise tracking under accurate models. Castano et al. [7] provide a benchmarking methodology for humanoid balancing controllers that introduces unified key performance indicators covering CoM displacement, recovery time, and ankle torque, but the evaluation is restricted to a single platform and a single control paradigm. Ramuzat et al. [8] benchmark several whole-body controllers on the TALOS humanoid robot and demonstrate that architectural differences

produce measurable performance divergence, but again restrict comparison to controllers within the same optimisation paradigm. No simulation-based study covers all four major WBC architectural classes — weighted QP, hierarchical QP, passivity-based, and reduced-order model-based — under a common, reproducible evaluation protocol on the same robot model.

A complementary limitation concerns the role of learning in augmenting model-based WBC. Residual reinforcement learning, introduced by Johannink et al. [9], demonstrated that decomposing a control problem into a model-based component and a learned residual correction improves both sample efficiency and final task performance relative to learning from scratch. The residual component is formulated as an additive torque correction that compensates for contact dynamics, friction, and modelling errors that are difficult to capture analytically, while the underlying model-based controller preserves physical consistency and constraint satisfaction. Kasaei et al. [10] demonstrated that combining a model-based gait kernel with a neural network residual substantially improves push-recovery performance compared to the model-based baseline alone. Hwangbo et al. [11] showed that carefully structured learning pipelines enable successful transfer of residual locomotion policies from simulation to hardware, motivating the simulation-based approach adopted in this thesis. However, no study has systematically evaluated how residual learning benefit varies across different base WBC architectures, or identified what training formulation properties are necessary for a learned residual to produce measurable improvement in a static balance context.

This thesis addresses both gaps. A simulation-based evaluation framework is developed for the Unitree H1 humanoid robot in MuJoCo [12], covering four WBC architectures under a common two-part disturbance protocol. The contribution is a structured comparative evaluation of the WBC design space, combined with a principled investigation of supervised residual learning augmentation across all four architectures.

1.2 Aim and Objectives

The primary aim of this thesis is to systematically evaluate and compare four WBC architectures for humanoid robot balance under controlled disturbance conditions in simulation, and to determine whether a learning-based residual correction layer can improve robustness beyond what model-based control alone achieves.

Three testable hypotheses guide the experimental evaluation. **H1:** Model-based QP-WBC exhibits a threshold-like, force-dependent degradation in stability performance, characterised by a sigmoidal fall-rate curve as a function of disturbance magnitude, consistent with the ankle torque saturation limits of the Unitree H1 platform. **H2:** The architectural distinction between weighted and hierarchical QP formulations produces equivalent stability performance when control objectives do not genuinely conflict, with measurable divergence emerging only when tasks genuinely compete for the same degrees of freedom — a condition verifiable empirically through a dedicated divergence test. **H3:** A history-augmented supervised residual MLP trained with a physically motivated behaviour cloning target provides statistically significant improvement in stability rate over the best model-based WBC baseline under medium-to-large disturbances, with the largest benefit occurring near the base controller's feasibility boundary.

The following objectives define the path to that aim. A reproducible simulation environment is established for the Unitree H1 humanoid in MuJoCo [12], incorporating a seeded stochastic disturbance system that generates genuinely independent, reproducible trials across varying force magnitudes, directions, body application sites, and push counts. Four WBC architectures — weighted QP, hierarchical QP, passivity-based, and LQR-on-LIPM

— are implemented under identical software infrastructure, enabling fair comparison across a shared evaluation protocol. A two-part disturbance evaluation protocol is designed and executed: a deterministic step-push grid for baseline characterisation, and a 200-seed stochastic multi-push batch for statistically meaningful comparative analysis following the benchmarking principles of Castano et al. [7]. A comprehensive metrics pipeline is developed recording CoM displacement, survival time, torque RMS, QP solve time, and a three-category outcome classification. Finally, a supervised residual MLP is designed, trained through a systematic ablation of three formulations, and evaluated against all four base controllers.

1.3 Research Questions

The investigation is guided by the following research questions.

Research Question 1: How do the four WBC architectures compare in terms of stability rate, CoM displacement, survival time, torque effort, and computational cost under a standardised stochastic disturbance protocol applied to the Unitree H1 in simulation?

Research Question 2: Under what disturbance conditions, if any, does the strict priority structure of hierarchical QP produce measurably different balance performance compared to weighted QP, and can this be verified empirically in a static balance push-recovery setting?

Research Question 3: To what extent does augmenting the best-performing model-based WBC with a supervised residual MLP improve balance robustness, and what training formulation properties are necessary for that improvement to emerge?

1.4 Contributions

Three contributions are made by this work.

The first is a four-architecture WBC comparative evaluation framework for the Unitree H1 in MuJoCo [12]. The framework covers weighted QP, hierarchical QP, passivity-based, and LQR-on-LIPM controllers under a common two-part disturbance protocol — a deterministic step-push grid and a 200-seed stochastic multi-push evaluation — producing 800 independent trials per controller across ten body application sites. A three-category outcome classification is introduced as part of this framework. Prior benchmarking work by Castano et al. [7] uses a binary stable/fell classification that cannot distinguish trials in which the robot survives but fails to return to the reference posture within the evaluation window. The third category — termed "neither" — is shown to differentially identify a class of lingering oscillation outcomes that separates architectural performance in a way binary classification cannot capture.

The second contribution is an empirical characterisation of the conditions under which weighted and hierarchical QP formulations produce equivalent or divergent outcomes. Across 200 independent stochastic trials, both controllers produce identical stability outcomes on every seed, with a mean per-trial CoM difference of 0.50 mm. The equivalence is shown to arise from the dominance of the balance objective weight, which means the strict priority structure of HQP is never exercised under these conditions. This finding is verified through a dedicated divergence test confirming the HQP implementation is correct, and is consistent with the theoretical prediction of Escande and Mansard [3] that HQP and WQP converge to the same solution when one task is sufficiently dominant.

The third contribution is a progressive ablation study of supervised residual learning applied to all four base controllers. Two formulations using a LIPM prediction loss produce null results, and the theoretical reason is identified: the LIPM's unstable eigenvalue causes

the loss gradient to be near-zero regardless of the correction applied, meaning the network learns to output zero corrections regardless of input. A third formulation incorporating five-frame temporal history, the base controller torque as a contextual input, and a pitch-rate behaviour cloning target produces a statistically significant improvement of 7.5 percentage points in stability rate over the passivity baseline, confirmed by McNemar's exact test at $p < 0.001$. This improvement generalises across all four base controllers and is largest in the 125–175 N force range, consistent with the observation of Kasaei et al. [10] that residual learning provides the greatest benefit at the boundary of the base controller's feasibility region.

1.5 Scope and Limitations

The scope of this work is restricted to static balance under push-recovery conditions: the Unitree H1 begins from a nominal standing posture and is evaluated on its ability to recover following one to five external push disturbances applied over a ten-second simulation window. Dynamic locomotion, stepping recovery, and manipulation tasks are explicitly outside the scope of this thesis. Stepping recovery would require a fundamentally different control architecture incorporating footstep planning and swing-leg trajectory generation, and is identified as the primary direction for future work.

The disturbance evaluation covers ten body sites — pelvis, torso, bilateral shoulder, elbow, hip, and knee — with force magnitudes sampled log-uniformly from 20 to 250 N. Forces above 250 N are excluded on the basis that all controllers converge to the same failure outcome above approximately 175 N due to ankle torque saturation at ± 40 Nm, making the region above 250 N uninformative for architectural comparison.

Several methodological limitations are explicitly acknowledged. All experiments are conducted in MuJoCo [12], whose contact and friction models, while high-fidelity, may not fully reproduce real-world hardware dynamics; no sim-to-real validation is performed, which is a recognised limitation in simulation-based control research [11]. Perfect state observation is assumed throughout, with no sensor noise or communication delays. The residual MLP is trained and evaluated on the same disturbance distribution; generalisation to out-of-distribution perturbations is not assessed. A single nominal starting posture is used for all trials, following standard push-recovery benchmarking practice [7], which means results characterise disturbance rejection from equilibrium rather than robustness across diverse initial conditions. Finally, the stochastic evaluation uses 200 seeds per controller, providing approximately 70% statistical power at $\alpha = 0.05$ for detecting the 5–6 percentage point differences observed, which is appropriate for a master's-level simulation study but leaves smaller effects undetected.

2 Related Work

2.1 Foundations of Whole-Body Control

The theoretical foundation for whole-body control of humanoid robots was established by Khatib [2] through the Operational Space Formulation (OSF). The central principle is that desired robot behaviour is specified directly in task space — as desired accelerations or forces at functionally meaningful locations such as the CoM or end effectors — and mapped to joint torques through the robot's dynamic model. For a robot with joint torque vector τ , joint position q , and task Jacobian J , the task-space control law takes the form:

$$\tau = J^T F_{\text{task}} + N^T \tau_{\text{posture}}$$

where F_{task} is the desired task-space force, J^T maps it to joint space, N is the null-space projector of the primary task, and τ_{posture} regulates secondary objectives in the remaining degrees of freedom. This formulation naturally handles kinematic redundancy: lower-priority tasks are projected into the null space of higher-priority tasks, enabling simultaneous satisfaction of multiple objectives. The OSF remains the theoretical basis for modern WBC regardless of whether the implementation uses analytical null-space projection or numerical optimisation.

The practical limitation of pure null-space projection is that it does not handle inequality constraints in a principled way. Real humanoid robots impose hard bounds on joint torques, contact forces must satisfy friction cone constraints to prevent slipping, and ground reaction forces must remain non-negative to maintain contact. Lee et al. [14] addressed this by reformulating the OSF as a constrained quadratic program (QP), enabling inequality constraint handling within the operational space framework while reducing the number of optimisation variables relative to prior approaches. A QP takes the general form:

$$\begin{aligned} &\text{minimise over } x: (1/2) x^T H x + c^T x \\ &\text{subject to: } A_{\text{eq}} x = b_{\text{eq}}, A_{\text{ineq}} x \leq b_{\text{ineq}} \end{aligned}$$

where H is a positive definite cost matrix, c is a linear cost vector, and the constraint matrices encode equality and inequality requirements. In WBC, x is the joint torque vector τ , H encodes the task tracking costs, and the constraints enforce torque limits and contact conditions. Lee et al. [14] demonstrated that this formulation can be solved at humanoid control frequencies, which is a prerequisite for any practical torque-controlled deployment.

A comprehensive survey of optimisation-based methods for dynamic legged robots is provided by Wensing et al. [13], who present a unified mathematical treatment covering OSF, QP-based WBC, hierarchical control, and model predictive control. Their framework situates the four architectures evaluated in this thesis within a common theoretical structure and identifies the conditions under which each architectural choice affects the feasibility and quality of the resulting torque solution.

2.2 Weighted and Hierarchical Quadratic Programming

The most widely adopted formulation of QP-based WBC combines all task objectives into a single weighted cost function solved simultaneously. For tasks $e_1, e_2, \dots, e_n$ with associated weight matrices $W_1, W_2, \dots, W_n$, the Weighted QP (WQP) optimisation is:

$$\begin{aligned} &\text{minimise over } \tau: \sum_i ||W_i e_i(\tau)||^2 \\ &\text{subject to: } \tau_{min} \leq \tau \leq \tau_{max} \end{aligned}$$

The WQP is computationally efficient and straightforward to implement, but it has a fundamental structural limitation: because all tasks compete in a shared weighted sum, the optimiser will sacrifice a high-priority task to partially satisfy a lower-priority one whenever the weighted cost of doing so is lower than fully satisfying both. For humanoid balance, this means that when balance and posture objectives conflict under large disturbances, the optimiser may partially violate the balance constraint if the posture cost is sufficiently large, producing a fall that a priority-enforcing architecture would have prevented.

Escande and Mansard [3] formalised the Hierarchical Quadratic Programming (HQP) framework as a principled solution to this problem. HQP arranges tasks in a strict lexicographic priority ordering and solves them sequentially. The first priority task is solved optimally:

$$\text{minimise over } \tau: ||e_1(\tau)||^2, \text{ subject to constraints}$$

The solution set of this first problem — the set of all τ that achieve the minimum value of $||e_1||^2$ — is then used as a constraint for the second priority problem:

$$\begin{aligned} &\text{minimise over } \tau: \\ &||e_2(\tau)||^2, \text{ subject to previous constraints and } ||e_1(\tau)||^2 = ||e_1^*||^2 \end{aligned}$$

where $||e_1^*||^2$ is the optimal value achieved in the first stage. This process continues for all priority levels. The key guarantee is that no lower-priority task can ever degrade a higher-priority one, because achieving the optimal first-priority value is enforced as a hard equality constraint in all subsequent stages. Escande and Mansard [3] demonstrated that their solver achieves a tenfold speed improvement over iterative null-space projection methods and operates at humanoid control frequencies, making lexicographic priority practically deployable.

An important theoretical relationship between WQP and HQP is that when one task is assigned a sufficiently dominant weight in the WQP — meaning the weight ratio between the highest and lowest priority tasks is large enough that the lower-priority task contributes negligibly to the optimal solution — both formulations converge to the same torque solution. This follows from the fact that in the limiting case of infinite balance weight, the WQP solution satisfies the balance constraint exactly, producing the same result as the HQP first-priority solution. Ramuzat et al. [8] confirmed this experimentally on the TALOS humanoid robot, demonstrating that WQP and HQP produce equivalent performance under non-conflicting task conditions. Measurable divergence emerges only when tasks genuinely compete for the same degrees of freedom, a condition that requires both tasks to make conflicting demands on the same joint subset simultaneously.

Herzog et al. [4] provided one of the earliest hardware demonstrations of hierarchical inverse dynamics WBC on a torque-controlled humanoid, integrating LQR-designed centroidal momentum control as the highest-priority task. Their work established that centroidal momentum — the combined linear and angular momentum of the full robot computed at the CoM — is the appropriate quantity to regulate for whole-body balance, because it captures the global effect of all joint motions on balance without requiring a simplified model. The centroidal momentum dynamics take the form:

$$[\dot{\mathbf{p}}; \dot{\mathbf{L}}] = \mathbf{A}(\mathbf{q}) \boldsymbol{\tau} + \mathbf{b}(\mathbf{q}, \dot{\mathbf{q}})$$

where $\mathbf{p}$ is linear momentum, $\mathbf{L}$ is angular momentum, $\mathbf{A}$ is the centroidal momentum matrix, and $\mathbf{b}$ captures gravitational and Coriolis effects. Regulating these quantities as the highest-priority task ensures that the balance objective is always satisfied first, with posture objectives resolved in the remaining null space.

2.3 Passivity-Based Whole-Body Control

An alternative to optimisation-based WBC is the passivity-based formulation introduced by Henze et al. [5]. Rather than minimising a cost function at each timestep, the passivity-based controller derives joint torques analytically from an energy-based Lyapunov function that guarantees the closed-loop system is passive. A system is passive if the stored energy at any time is bounded by the energy supplied externally minus the energy dissipated internally, expressed as:

$$V(\mathbf{x}) \leq -\alpha \|\mathbf{x}\|^2 + \mathbf{u}^T \mathbf{y}$$

where V is a Lyapunov function representing stored energy, $\mathbf{x}$ is the state, $\mathbf{u}$ is the input, $\mathbf{y}$ is the output, and $\alpha > 0$ ensures energy dissipation. For a humanoid balance controller, this condition guarantees that the controller cannot inject energy into the system: disturbance-induced kinetic energy is always dissipated, preventing spontaneous acceleration away from equilibrium.

The passivity-based WBC of Henze et al. [5] achieves this through a damping injection term in the torque command:

$$\boldsymbol{\tau} = \boldsymbol{\tau}_{\text{gravity}} + \boldsymbol{\tau}_{\text{PD}} - \mathbf{b} \cdot \dot{\mathbf{q}} + \boldsymbol{\tau}_{\text{balance}}$$

where $\boldsymbol{\tau}_{\text{gravity}}$ compensates gravitational and Coriolis forces, $\boldsymbol{\tau}_{\text{PD}}$ is a proportional-derivative posture term, the term $-\mathbf{b} \cdot \dot{\mathbf{q}}$ with positive damping gain $\mathbf{b}$ is the passivity-enforcing damping injection, and $\boldsymbol{\tau}_{\text{balance}}$ provides CoM regulation. The damping injection term ensures that $V \leq -\mathbf{b} \cdot \dot{\mathbf{q}} \cdot \dot{\mathbf{q}} < 0$ whenever the robot is moving, meaning the controller dissipates kinetic energy at every timestep. This property produces intrinsically compliant, energy-dissipating responses to disturbances, which is behaviourally distinct from the cost-minimising but not necessarily energy-dissipating responses of QP-WBC.

Sovukluk et al. [6] conducted an experimental comparison between inverse-dynamics WBC and passivity-based WBC on a shared hardware platform under standardised perturbation conditions. Their central finding was that the passivity formulation exhibits superior robustness to modelling errors because the energy-dissipating structure provides a stability margin that does not depend on model accuracy, while inverse-dynamics WBC achieves more precise tracking under accurate models because it exploits the full dynamic model to

compute optimal torques. This performance tradeoff — compliance and robustness versus optimality and precision — is a central theme of the comparative evaluation conducted in this thesis.

2.4 Reduced-Order Balance Models and LQR Control

Full-body WBC requires solving a QP over the complete joint space at every control step, which carries significant computational cost. A simpler alternative treats the humanoid as a reduced-order model for the purpose of computing balance torques, sacrificing full-body dynamic consistency in favour of analytical tractability.

The Linear Inverted Pendulum Model (LIPM) is the most widely used reduced-order model for humanoid balance [16]. The LIPM reduces the full-body dynamics to a point mass m constrained to move at constant height z_c above the ground, supported by a massless leg. The resulting CoM dynamics in the sagittal plane are:

$$\ddot{x}_{CoM} = (g / z_c) \cdot (x_{CoM} - x_{CoP})$$

where x_{CoM} is the horizontal CoM position, x_{CoP} is the centre of pressure, and g is gravitational acceleration. This equation is linear, enabling analytical solutions and real-time planning. The LIPM has been used extensively for ZMP-based gait generation and CoM trajectory planning as reviewed by Elhasairi and Pechev [16], though its constant-height assumption is violated during large disturbances when the robot's CoM height changes significantly.

The capture point, introduced by Pratt et al. [15], is derived analytically from the LIPM. It is defined as the unique ground position to which the robot must step to bring its CoM velocity to zero without further control input:

$$x_{CP} = x_{CoM} + \dot{x}_{CoM} / \omega_0$$

where $\omega_0 = \sqrt{g / z_c}$ is the natural frequency of the LIPM. The capture point provides a criterion for distinguishing ankle-strategy recovery — where the CoP can be placed within the support polygon to arrest the fall — from stepping-strategy recovery, where it cannot. This criterion is directly relevant to the experimental design of this thesis, as the evaluation is restricted to conditions where ankle-strategy recovery is at least theoretically possible.

Stephens [17] demonstrated that a Linear Quadratic Regulator (LQR) applied to the discretised LIPM produces an effective balance controller for standing humanoids. The LQR minimises the infinite-horizon quadratic cost:

$$J = \sum (x^T Q x + u^T R u)$$

where $x = [x_{CoM}, \dot{x}_{CoM}, y_{CoM}, \dot{y}_{CoM}]^T$ is the four-dimensional CoM state, $u = [x_{CoP}, y_{CoP}]^T$ is the commanded centre of pressure, and Q and R are positive definite weighting matrices. The resulting optimal gain matrix K maps CoM state error to CoP correction. The ankle torque command is then derived from the CoP correction via the static relationship $\tau_{ankle} = m \cdot g \cdot CoP_{cmd}$, which follows from the moment equilibrium condition at the ankle joint when the foot is flat on the ground. Stephens [17] validated this approach on a physical humanoid, demonstrating recovery from disturbances of up to 17 Ns impulse.

2.5 Push Recovery Evaluation and Benchmarking

Push recovery is the standard experimental test of humanoid balance robustness. The evaluation methodology has evolved from qualitative observation of single pushes toward quantitative, multi-condition protocols with standardised metrics. Castano et al. [7] provide the most systematic existing methodology, using the EUROBENCH framework to compare three balance strategies on the REEM-C humanoid under standardised push conditions. The key performance indicators introduced in that study — CoM deviation, settling time, ankle torque RMS, and reconstructed centre of pressure — have been adopted as the standard metrics for humanoid balance benchmarking and directly inform the evaluation framework developed in this thesis.

Heng et al. [20] extended push-recovery evaluation to multi-directional disturbances and uneven terrain, using a momentum-based observer to estimate disturbance forces without dedicated force sensors. Their evaluation on the BRUCE bipedal robot under upper-body pushes of varying magnitude and direction provides a methodological precedent for the multi-directional stochastic evaluation adopted in Part 1b of this thesis. Zhang et al. [21] demonstrated that combining predictive control with WBC improves push recovery robustness over pure WBC in simulation, confirming that the performance gap between model-based WBC and an enhanced controller is experimentally detectable in simulation studies. Kim et al. [18] evaluated an MPC-based capture point controller under pelvis pushes of up to 150 N, providing quantitative grounding for the force range selected in the present evaluation.

A shared limitation of all existing push-recovery benchmarking studies is the use of deterministic, fixed-parameter disturbance protocols. When the same force vector is applied to the same initial state, every trial produces exactly the same trajectory, meaning repeated trials within a condition are not statistically independent and standard hypothesis testing is not valid. This limitation is explicitly acknowledged in the present thesis, and the Part 1b stochastic evaluation is designed to address it by generating each trial from a unique random seed, ensuring genuine statistical independence as advocated by the standardised evaluation principles of Sferrazza et al. [27].

2.6 Learning-Based Residual Compensation

Model-based WBC performance is bounded by the accuracy of the robot model on which it is based. Contact dynamics, joint friction, actuator compliance, and unmodelled flexible modes introduce systematic discrepancies between model-predicted and actual robot behaviour that degrade controller performance under realistic conditions. Learning-based residual compensation addresses this limitation by augmenting the model-based controller with a learned additive correction.

Johannink et al. [9] introduced residual reinforcement learning as a formal framework for this decomposition. The total control command is expressed as:

$$\tau_{total} = \tau_{WBC} + \delta\tau_{RL}$$

where τ_{WBC} is the model-based controller output and $\delta\tau_{RL}$ is a residual correction learned through reinforcement learning. The RL agent receives a reward based on task performance and learns to produce corrections that compensate for the gap between the model-based controller and optimal performance. The authors demonstrated that this approach achieves higher sample efficiency than learning from scratch, because the model-based component provides a near-feasible initialisation that keeps the residual correction small and the

exploration problem tractable. The decomposition principle is adopted directly in this thesis, with supervised behaviour cloning used in place of reinforcement learning for reasons discussed in Chapter 5.

Kasaei et al. [10] demonstrated the effectiveness of residual augmentation for push-recovery in a locomotion setting. Their finding that the largest improvements occur in the medium-force regime — where the base controller operates near its feasibility boundary — provides the experimental prediction that residual learning will be most beneficial in the 125–175 N force range for the controllers evaluated here. Hwangbo et al. [11] demonstrated that carefully structured neural network learning pipelines enable successful transfer of locomotion policies from simulation to hardware, establishing MuJoCo as a valid training environment for learning-based augmentation and providing evidence that simulation-trained residuals can generalise to physical systems.

Yan et al. [22] formalised the I-CTRL framework, which constrains residual reinforcement learning within bounded deviations from a reference trajectory to maintain physical feasibility while improving task performance. Their use of bounded output correction directly informs the ± 20 Nm per joint output bound adopted for the residual MLP in this thesis. Wang and Huang [23] demonstrated that hierarchical WBC combined with learned motion primitives outperforms either component alone in dynamic gait generation, providing further evidence for the complementarity of model-based and learning-based control.

The supervised learning variant of residual compensation — in which the correction is learned from labelled state-torque pairs rather than from RL rewards — has been explored in manipulation contexts but not systematically in humanoid balance. The distinction between supervised residual learning and residual RL is important: supervised learning requires labelled ground-truth corrections, which are not directly observable when the true optimal torque is unknown. This thesis investigates three supervised formulations of increasing sophistication and characterises the conditions under which the labelling problem can be resolved through physically motivated proxy targets.

2.7 Identified Research Gaps

The review of related work identifies three gaps that this thesis is positioned to address.

The first gap is the absence of a simulation-based comparative evaluation covering all four major WBC architectural classes under a common, reproducible disturbance protocol on the same robot model. Sovukluk et al. [6] provide the closest existing comparison but are restricted to hardware, to two architectures, and to a deterministic protocol. Ramuzat et al. [8] cover multiple controllers on the TALOS platform but within a single optimisation paradigm. Castano et al. [7] provide a benchmarking methodology but apply it to a single architecture. A simulation study covering all four classes under a stochastic evaluation protocol does not exist in the literature.

The second gap is the absence of a three-category outcome classification in push-recovery benchmarking. All existing studies use a binary stable/fell classification. A robot can survive the evaluation window without recovering to the reference posture — a distinct outcome that the binary classification conflates with successful recovery. This category is shown in this thesis to differentially characterise the energy-dissipating behaviour of the passivity controller.

The third gap is the systematic characterisation of residual learning benefit across multiple base controllers of different architectural types. Johannink et al. [9] and Kasaei et al. [10] each evaluate a single base controller. No study has applied the same residual learning framework to controllers of fundamentally different types and characterised how improvement

magnitude depends on the structural properties of the base controller. This thesis provides that characterisation across four controllers.

3 Method

3.1 Research Approach

The methodology adopted in this thesis follows a four-phase structure designed to progress systematically from controller implementation through comparative evaluation to learning-based augmentation. The phases are executed sequentially so that each phase builds on validated outputs from the previous one, ensuring that comparative conclusions rest on individually verified components rather than on simultaneous untested assumptions.

In the first phase, a simulation environment is established for the Unitree H1 humanoid in MuJoCo [12], and four WBC architectures are implemented and individually validated under zero-disturbance standing conditions. Validation at this stage confirms that each controller achieves stable standing before any disturbance experiment is conducted, isolating implementation correctness from disturbance response. In the second phase, a deterministic step-push grid experiment (Part 1a) is executed to characterise the baseline performance envelope of each controller across fixed force levels and body sites. This phase provides qualitative understanding of each controller's failure mode transitions before the statistically rigorous evaluation. In the third phase, a stochastic multi-push evaluation (Part 1b) is conducted across 200 independent seeded trials per controller, producing the primary dataset from which all quantitative comparisons are drawn. The stochastic design is adopted specifically because deterministic trial repetition does not produce statistically independent samples, as discussed in Section 2.5. In the fourth phase, a supervised residual MLP is designed, trained through a progressive ablation of three formulations, and evaluated against all four base controllers using the same 200-seed protocol.

This four-phase structure reflects standard practice in simulation-based WBC evaluation established by Castano et al. [7] and extended by Ramuzat et al. [8], and ensures that experimental findings are traceable to specific design decisions.

3.2 Simulation Environment and Robot Platform

3.2.1 MuJoCo Physics Engine

All experiments are conducted in MuJoCo version 3.x [12], a physics engine specifically designed for model-based robot control. MuJoCo uses a convex optimisation formulation for contact force resolution, producing physically consistent constraint forces without the penetration artefacts common in penalty-based simulators. This property is important for the standing balance scenario evaluated here, where foot-ground contact forces must be accurately computed at every timestep to produce correct joint torque responses from the WBC controllers described in Chapter 2.

The simulation timestep is set programmatically to 1 ms (1000 Hz) to ensure stable contact dynamics, while the WBC control loop executes at 100 Hz — every tenth simulation step. This two-level frequency separation reflects standard practice in humanoid simulation [12] and ensures that contact events are fully resolved before each new torque command is computed. All data logging is performed at the control rate of 100 Hz and saved to CSV format for offline analysis.

3.2.2 Unitree H1 Robot Model

The Unitree H1 is selected as the experimental platform because it is a full human-scale, torque-controlled humanoid whose dynamics directly reflect the balance challenges described in Chapter 1. With a standing height of approximately 1.8 m and a total mass of 47 kg, the H1 produces the high centre of mass, narrow support polygon, and large gravitational moments that make whole-body balance a non-trivial control problem requiring the coordinated multi-joint strategies described in Chapter 2. The robot model is sourced from the MuJoCo Menagerie repository, a community-maintained collection of validated MJCF robot models [26].

The H1 has 19 actuated degrees of freedom distributed across the lower body and upper body. The ankle pitch joints, limited to ± 40 Nm, constitute the binding constraint in the balance experiments conducted in this thesis. As established by the capture point framework of Pratt et al. [15] and described in Section 2.4, the ankle-strategy recovery regime is bounded by the ability of the ankle joints to produce sufficient CoP correction. For the Unitree H1 at its nominal standing height, this limit is reached at approximately 100–150 N of applied disturbance force, above which ankle saturation prevents further corrective torque regardless of the WBC architecture employed. All actuator gear ratios are set to 1.0 in the MJCF model, meaning commanded torques map directly to joint torques without transmission scaling. The actuator layout and torque limits relevant to balance control are summarised in Table 3.1.

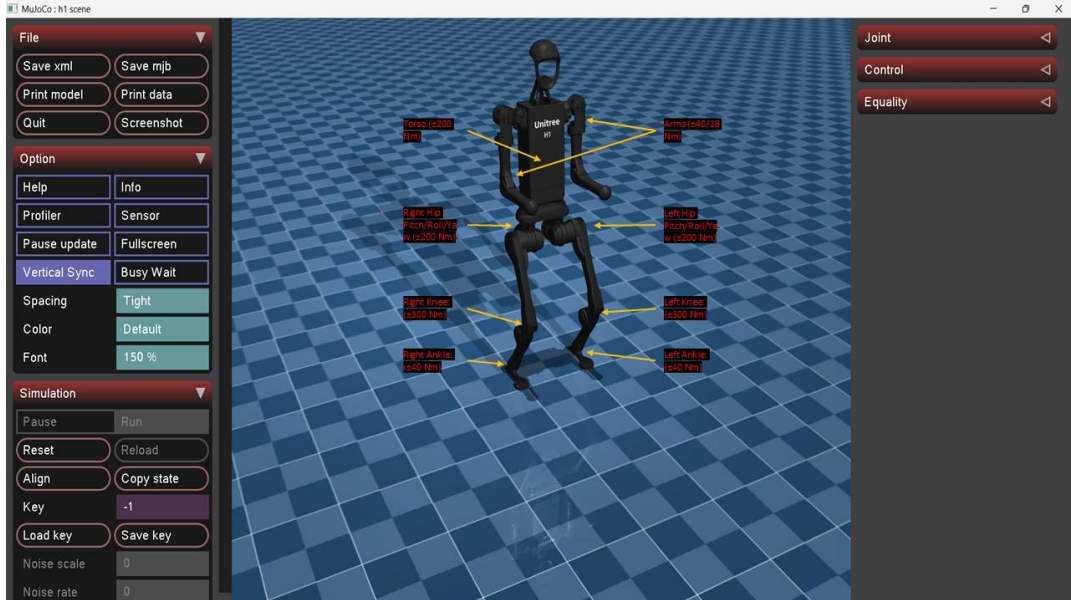


Figure 3.1 Unitree H1 humanoid robot joint layout showing the 19 actuated degrees of freedom and corresponding torque limits. Ankle pitch joints (± 40 Nm, highlighted) constitute the binding constraint in the balance experiments. Hip, knee, and torso joints operate at ± 200 – 300 Nm.

Table 3.1 Unitree H1 actuator layout and torque limits relevant to balance control.

Index	Joint	Torques Limit(Nm)
0	Left hip yaw	± 200
1	Left hip roll	± 200
2	Left hip pitch	± 200
3	Left knee	± 300
4	Left ankle pitch	± 40
5	Right hip yaw	± 200
6	Right hip roll	± 200
7	Right hip pitch	± 200
8	Right knee	± 300
9	Right ankle pitch	± 40
10	Torso	± 200
11	Left shoulder pitch	± 40
12	Left shoulder roll / yaw / elbow	± 40 / ± 18
13	Right shoulder pitch	± 40
14	Right shoulder roll / yaw / elbow	± 40 / ± 18

The home keyframe defined in the MJCF model places the pelvis at a height of 0.935 m with knees flexed at 0.8 rad. This configuration is used as the initial and reference posture for all trials. All controllers are initialised from this keyframe, and all recovery metrics are computed relative to the CoM position corresponding to this posture.

3.3 Controller Implementation

Four WBC architectures are implemented in Python using the MuJoCo Python API. All four share identical infrastructure: the same MJCF model, the same control loop timing, the same fall detection logic, and the same metrics computation pipeline. Only the torque computation method differs between controllers. This design ensures that observed performance differences are attributable to architectural choices rather than implementation inconsistencies, which is the fundamental requirement for a fair comparative evaluation as described by Castano et al. [7].

A validation procedure is applied to each controller before any disturbance experiment is conducted. Validation criteria are: stable standing for at least 10 seconds with no fall detection trigger, pelvis height remaining above 0.80 m throughout, and torso pitch remaining within ± 0.1 rad. Controllers that fail these criteria are debugged and corrected before proceeding. This staged validation approach ensures that disturbance response results reflect genuine architectural differences rather than implementation defects.

3.3.1 Weighted Quadratic Program Controller

The Weighted QP controller implements the formulation described in Section 2.2, solving a single quadratic program at each 100 Hz control step. The optimisation minimises a weighted sum of balance and posture task errors over the joint torque vector τ :

$$\begin{aligned}
 &\text{minimise over } \tau: \quad w_p \cdot \|e_{pitch}(\tau)\|^2 + w_r \cdot \|e_{roll}(\tau)\|^2 + w_q \cdot \\
 &\quad \|e_{posture}(\tau)\|^2 + w_{reg} \cdot \|\tau\|^2 \\
 &\text{subject to: } \tau_{min} \leq \tau \leq \tau_{max}
 \end{aligned}$$

where e_{pitch} and e_{roll} are sagittal and frontal plane balance errors, $e_{posture}$ is the joint angle deviation from the nominal standing configuration, and the regularisation term prevents excessive actuation. The balance error signals follow the ankle strategy formulation of Stephens [17], as presented in Section 2.4:

$$\begin{aligned} e_{pitch} &= k_p \cdot (\theta_{pitch} - \theta_{ref}) + k_d \cdot \dot{\theta}_{pitch} \\ e_{roll} &= k_p \cdot \theta_{roll} + k_d \cdot \dot{\theta}_{roll} + k_i \cdot \int \theta_{roll} dt \end{aligned}$$

Inverse dynamics compensation is incorporated by adding MuJoCo's `qfric_bias` vector — which contains gravity and Coriolis forces — to the QP solution at each step, following the approach of Herzog et al. [4]. This means the QP optimises corrective torques relative to the gravity-compensated baseline rather than absolute torques, which reduces the required optimisation range and improves numerical conditioning.

The balance weight w_p is set to 2000 and the posture weight w_q to 1.0, producing a 2000:1 ratio. As established in Section 2.2, this ratio is sufficient to make the balance objective effectively dominant, meaning the WQP and HQP solutions should converge to the same result when tasks do not conflict. The QP is solved using the CVXPY interface with the OSQP backend. The mean solve time is 0.84 ms per step, well within the 10 ms budget available at 100 Hz.

3.3.2 Hierarchical Quadratic Program Controller

The Hierarchical QP controller implements the two-stage priority structure of Escande and Mansard [3], as described in Section 2.2. Balance is enforced as the strict first-priority task and posture regulation is optimised as the second-priority task only within the solution space of the first stage.

In Stage 1, a balance QP minimises sagittal and frontal balance errors on the ankle pitch and hip pitch joints. Hip roll is regulated via a separate integral feedback term appended to the Stage 1 objective. The Stage 1 solution $\tau_{balance}$ is balance-optimal. In Stage 2, a posture QP minimises joint angle deviation from the nominal posture, subject to a soft near-equality penalty on the ankle pitch and hip pitch torques with respect to $\tau_{balance}$. The soft penalty weight is set to 5000, making the constraint effectively binding in normal operating conditions.

The choice of soft penalty rather than hard equality constraints in Stage 2 is justified by the numerical stability analysis of Wensing et al. [13]: hard equality constraints in Stage 2 produce solver infeasibility when the Stage 1 solution approaches torque limits, causing the entire Stage 2 solve to fail. The soft penalty formulation degrades gracefully in these conditions. Knee and hip roll joints are left free in Stage 2, allowing posture regulation on these joints without constraining the balance-critical ankle and hip pitch strategy.

The mean total solve time is 1.58 ms per step — 1.89 times the cost of the single-stage WQP — consistent with the increased computational cost expected from the two-stage architecture. The HQP implementation is verified through a dedicated divergence test described in Section 3.3.5.

3.3.3 Passivity-Based Controller

The passivity-based controller implements the energy-shaping control principle of Henze et al. [5], as described in Section 2.3. The torque command is assembled as an additive four-layer computation without a QP solver:

$$\tau = \tau_{gravity} + \tau_{posture} + \tau_{damping} + \tau_{balance}$$

where τ_{gravity} compensates gravitational and Coriolis forces via MuJoCo's `qfric_bias`, τ_{posture} is a PD term regulating joint angles to the nominal standing configuration, $\tau_{\text{damping}} = -b \cdot \dot{q}$ is the passivity-enforcing damping injection with gain $b = 20.0$ Nm·s/rad, and τ_{balance} is a pitch-rate and roll-rate stabilising correction applied to the balance-critical joints.

As established in Section 2.3, the passivity condition is guaranteed by the damping injection layer: the term $-b \cdot \dot{q}$ ensures that the controller's contribution to the total energy time derivative is non-positive for any positive b , meaning the controller cannot inject energy into the system. This property distinguishes the passivity controller from the QP-based controllers architecturally: QP controllers minimise a cost function but do not explicitly guarantee energy dissipation, while the passivity controller guarantees energy dissipation but does not minimise any explicit cost. The damping gain $b = 20.0$ is selected through iterative validation testing to produce stable standing with smooth disturbance responses without excessive joint braking during nominal motion.

The passivity controller is selected as the base controller for the residual MLP evaluation, justified by three properties. First, its algebraic torque computation produces smooth, structured outputs that are more amenable to supervised learning than the discontinuous outputs of QP-based controllers near constraint boundaries. Second, as established by Sovukluk et al. [6] and discussed in Section 2.3, the passivity formulation exhibits superior robustness to modelling errors, meaning its systematic errors arise from structured model-plant mismatch rather than optimisation artefacts, and are therefore more learnable. Third, the passivity controller produces zero QP solve overhead, meaning the residual MLP adds computational cost without competing with an existing solver budget.

3.3.4 LQR-on-LIPM Controller

The LQR-on-LIPM controller implements the reduced-order balance formulation of Stephens [17], as described in Section 2.4. The LIPM dynamics are discretised at the simulation timestep with CoM height $z_c = 0.9504$ m, corresponding to the home keyframe CoM height, and the LQR gain matrix K is computed once at initialisation using the discrete algebraic Riccati equation via SciPy's `linalg.solve_discrete_are` function. The resulting closed-loop eigenvalues are 0.997 and 0.968, confirming stability of the discrete-time closed-loop system.

The LQR gain maps the four-dimensional CoM state — horizontal position and velocity in the sagittal and frontal planes — to a commanded centre of pressure correction. This CoP correction is converted to ankle pitch torques via the static moment equilibrium relationship $\tau_{\text{ankle}} = m \cdot g \cdot \text{CoP_cmd}$ derived by Stephens [17], where m is the robot total mass and g is gravitational acceleration. A posture PD layer handles non-balance joints — knees, hip yaw, torso, and arms — and a separate torso orientation regulator acts on the hip pitch joints to maintain trunk orientation. The LQR gain is $K = [-10.871, -3.428]$ for the sagittal plane, scaled symmetrically for the frontal plane with an empirical calibration coefficient of 0.1 validated through standing trials.

The LQR-on-LIPM controller is included in this evaluation because, as established in Section 2.4, it represents the class of reduced-order balance controller underlying many practical humanoid deployments. Its inclusion quantifies the performance cost of simplifying from full-body dynamics to a CoM-only model, providing a scientifically informative reference point for understanding when the additional computational cost of QP-based WBC is justified.

3.3.5 Implementation Verification and Pitfalls

Several implementation issues were encountered during controller development that are informative for understanding the practical challenges of WBC implementation beyond what the theoretical literature describes.

The first concerns sign conventions in the balance correction terms. The direction of the balance correction — whether positive pitch error requires positive or negative ankle torque — depends on the MJCF coordinate frame convention, which differs between robot models and cannot be assumed from theory. For the Unitree H1, the correct sign is determined empirically through a calibration procedure that applies a small perturbation and observes the correction direction. Applying an incorrect sign produces a controller that actively amplifies disturbances rather than correcting them. Critically, this failure mode is undetectable from the QP solve status, which returns optimal regardless, because the QP problem is correctly formulated — the error is in the physical interpretation of its parameters. This observation is consistent with the implementation verification principles discussed by Herzog et al. [4], who note that correct sign conventions in WBC must be verified through physical reasoning and diagnostic logging rather than assumed from formulation alone.

The second concerns derivative initialisation in discrete-time PD controllers. Initialising the previous pitch angle to zero at trial start rather than to the actual robot state produces a large derivative spike at the first control step, because the estimated pitch rate is computed as $(\theta_{\text{current}} - 0) / dt$ rather than the true rate. For the balance gains used here, this spike can exceed 1000 Nm and destabilise the robot within the first simulation step. The QP returns optimal during this destabilisation because the problem formulation is correct — the physical error is in the initialisation of the state estimate, not the solver.

The third concerns CVXPY disciplined parameterised programming compliance. CVXPY requires that QP problems be DPP-compliant to enable warm-starting and efficient re-solving at each timestep. Multiplying two CVXPY parameter objects inside the QP objective violates DPP compliance and produces a significant computational penalty at each solve. All parameter products must be precomputed in Python before being passed to the QP objective, which is not obvious from standard CVXPY usage examples.

The HQP implementation is verified through a dedicated divergence test in which the balance and posture task weights are set to equal values. Under equal weighting, the WQP and HQP solutions diverge by up to 200 Nm per joint at the hip roll joints, confirming that the hierarchical priority structure is correctly implemented and active. The equivalence observed in the main experiments therefore reflects the dominant balance weight ratio rather than an implementation error.

3.4 Disturbance Injection Framework

3.4.1 Part 1a: Deterministic Step-Push Grid

External push forces are applied to the robot through MuJoCo's `xfrc_applied` field, which adds a force and torque vector directly to a named rigid body at each simulation step for a specified duration. Forces are applied as impulses of 0.1 seconds duration at $t = 3.0$ seconds into each 10-second trial, giving the robot 3 seconds to reach steady state before the disturbance and 7 seconds to recover.

The Part 1a disturbance parameter space covers six force magnitudes (50, 100, 150, 200, 250, and 300 N), two body locations (pelvis and torso), and three directions (forward, backward, and lateral). Each condition is repeated across five trials with small random joint position perturbations applied at initialisation. This gives $6 \times 2 \times 3 \times 5 = 180$ trials per controller.

The force range of 50–300 N spans the full recovery regime for the Unitree H1, from the light disturbances at which all controllers succeed to the large disturbances at which all controllers fail due to ankle torque saturation.

As discussed in Section 2.5, a fundamental limitation of this design is that each combination of force, direction, and body site produces a deterministic simulation: the same force vector applied to the same initial state yields the same trajectory every trial. The five trials within each condition differ only through small initialisation noise, which is insufficient to produce statistically independent samples. Part 1a is therefore retained solely as a baseline characterisation tool that reveals the qualitative shape of each controller's force-response curve. All quantitative statistical comparisons are drawn exclusively from the Part 1b stochastic evaluation.

3.4.2 Part 1b: Stochastic Multi-Push Evaluation

The Part 1b experiment addresses the statistical independence limitation of Part 1a by generating each trial from a unique integer seed that fully determines the disturbance profile for that trial. The same seed always produces the same profile, ensuring reproducibility, while different seeds produce genuinely independent disturbance scenarios. This approach follows the reproducible evaluation principles of Sferrazza et al. [27] and directly addresses the deterministic protocol limitation identified in the benchmarking literature discussed in Section 2.5.

For each trial, the seed determines the following quantities. The number of pushes is drawn from a Poisson distribution with mean 2.5, clamped to [1, 5], reflecting that realistic disturbance sequences involve multiple contacts rather than a single isolated push. Individual push force magnitudes are sampled log-uniformly from 20 to 250 N. The log-uniform distribution is chosen because it assigns equal probability mass per decade of force, ensuring proportional coverage across the full force range without over-representing either very light or very heavy pushes. The upper bound of 250 N is justified by the ankle saturation analysis in Section 3.2.2: above approximately 175 N, all controllers converge to the same failure outcome, making forces above 250 N uninformative for architectural comparison. Push duration is sampled uniformly from 0.05 to 0.20 seconds, spanning the impulse contact range reported by Heng et al. [20]. Push direction is sampled uniformly in azimuth with an elevation tilt of up to ± 15 degrees.

The weighting reflects the mechanical significance of each site: pelvis and torso together account for 40% of sampling probability because they are the highest-mass sites whose perturbation most directly challenges CoM regulation, as established by the CoM-centric balance theory of Section 2.1. Forces applied to arm and lower-limb sites are scaled by factors of 0.60 and 0.80 respectively to reflect their reduced mechanical leverage relative to the central body mass. The push timing is constrained so that the first push occurs between $t = 1.0$ and $t = 2.5$ seconds, subsequent pushes are separated by at least 1.5 seconds, and all pushes complete before $t = 8.5$ seconds. The body site for each push is sampled from a weighted categorical distribution across ten body sites, as specified in Table 3.2.

Table 3.2 Part 1b body site sampling weights and force scaling factors.

Body site	Sampling weight	Force scale
Pelvis	0.20	1.00
Torso link	0.20	1.00
Left shoulder roll	0.08	0.60
Right shoulder roll	0.08	0.60
Left elbow	0.08	0.60
Right elbow	0.08	0.60
Left hip pitch	0.08	0.80
Right hip pitch	0.08	0.80
Left knee	0.06	0.60
Right knee	0.06	0.60

Two hundred seeds per controller are evaluated, producing 800 total independent trials across all four controllers. At the observed stability differences of 5–7 percentage points between controllers, 200 seeds provides approximately 70% statistical power at $\alpha = 0.05$, which is appropriate for a master's-level simulation study while remaining computationally feasible with sequential evaluation on standard hardware.

3.4.3 Extended Evaluation: Lower-Body Push Sites

Following the primary Part 1b evaluation, an extended dataset is collected using an updated disturbance generator incorporating all ten body sites described in Table 3.2, including the four lower-body sites that were absent from an earlier version of the evaluation. This extended dataset uses the same 200 seeds as the primary evaluation but produces different disturbance profiles because the body site distribution changes. The purpose of this extension, recommended during supervisory review, is to assess whether the primary findings generalise to lower-body disturbances, which test direct joint loading rather than CoM-dominant perturbation and represent a qualitatively different mechanical challenge. Results from the extended evaluation are reported separately in Chapter 4 to avoid conflating the two experimental conditions.

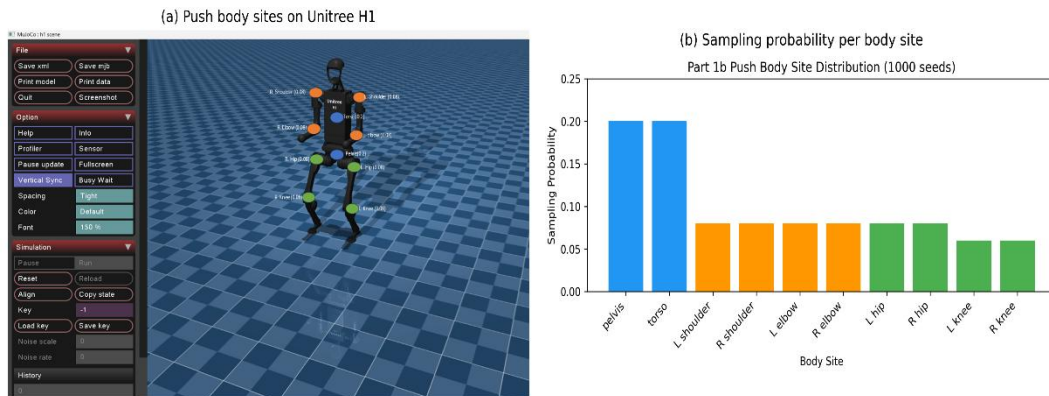


Figure 3.2 Part 1b disturbance body site distribution. **Left:** the ten push application sites on the Unitree H1, marked by sampling weight category — central body sites (pelvis and torso, weight 0.20 each), arm sites (shoulders and elbows, weight 0.08 each), and lower-limb sites (hip pitch and knee, weight 0.06–0.08). See figure legend for category key. **Right:** sampling probability per body site as specified in Table 3.2, verified across 1000 seeds.

3.5 Outcome Classification and Performance Metrics

3.5.1 Three-Category Outcome Classification

Each trial is classified into one of three outcome categories at the end of the 10-second evaluation window. A trial is classified as **Stable** if the robot's CoM returns to within 20 mm of the reference position and the pelvis height remains above 0.65 m throughout the trial. A trial is classified as **Fell** if the pelvis height drops below 0.65 m or the torso pitch exceeds 1.2 rad at any point during the trial, at which point the fall time is recorded and the trial is terminated. A trial is classified as **Neither** if neither of the above conditions is met at the end of the evaluation window: the robot has not fallen but has also not returned to the reference posture.

The introduction of the Neither category is grounded in nonlinear control theory and is motivated by a specific limitation of the binary stable/fell classification used in all existing push-recovery benchmarking studies reviewed in Section 2.5. In nonlinear dynamical systems, multiple equilibria can coexist: a system can be Lyapunov stable — meaning it remains bounded near some equilibrium — without being asymptotically stable — meaning it does not necessarily return to the original equilibrium [13]. A humanoid robot under a large disturbance can settle into a locally stable but displaced quasi-equilibrium that satisfies the controller's internal cost function approximately without returning to the nominal standing posture. This behaviour is theoretically expected for both QP-based controllers, which minimise a cost function without guaranteeing asymptotic convergence to the reference posture, and for the passivity-based controller, which guarantees energy dissipation but not convergence to a specific configuration. The Neither category captures exactly this class of outcome: the robot is physically stable in the Lyapunov sense — it has not fallen — but has not achieved asymptotic recovery to the reference posture within the evaluation window.

This distinction is practically important because binary classification conflates Neither trials with successful recovery, artificially inflating the stable rate and masking architectural differences. As demonstrated in the results of Chapter 4, the Neither category disproportionately affects the QP-based controllers — 14 trials each, representing 7.0% — compared to the passivity-based controller — 9 trials, representing 4.5%. This difference is attributable to the energy-dissipating property of the passivity controller's damping injection layer, which actively removes residual kinetic energy even during low-amplitude oscillation near the equilibrium, resolving the lingering oscillatory states that the QP controllers cannot escape. A binary classification framework cannot reveal this architectural difference because it does not distinguish between full recovery and mere survival. The three-category framework therefore represents a methodological contribution to humanoid balance benchmarking that extends the binary framework of Castano et al. [7] to the fixed-stance long-window evaluation context in which lingering quasi-equilibria are physically possible.

3.5.2 Quantitative Performance Metrics

The following metrics are computed for each trial and recorded in the results CSV, consistent with the KPI framework of Castano et al. [7].

- **Survival time** (s) is defined as $\min(\text{fall_time}, \text{simulation_end_time})$ and is always defined regardless of outcome category. For Stable and Neither trials, survival time equals 10.0 s. For Fell trials, it equals the recorded fall time. Survival time serves as the primary continuous stability metric because, unlike recovery time, it is defined for all three outcome categories.

- **Maximum CoM displacement** (m) is the maximum horizontal Euclidean distance between the instantaneous CoM position and the pre-disturbance reference CoM position over the full trial duration.
- **RMS CoM displacement** (m) is the root-mean-square of the horizontal CoM displacement integrated from the first push time to the end of the trial, providing a measure of the average response severity rather than the single worst instant.
- **Recovery time** (s) is defined only for Stable trials and is the elapsed time from the end of the last scheduled push to the first instant at which CoM displacement falls below the 20 mm threshold and remains below it for at least 80% of a subsequent 0.5-second sliding window. The sliding window criterion prevents brief threshold crossings from being recorded as recovery.
- **Torque RMS** (Nm) is the root-mean-square of all joint torques over the trial duration, decomposed into ankle, knee, and hip group contributions, measuring total control effort.
- **QP solve time** (ms) is recorded as mean and maximum per trial for the WQP and HQP controllers. Passivity and LQR controllers record zero for this metric, reflecting their zero solver overhead.

3.5.3 Statistical Analysis

Pairwise controller comparisons on binary stability outcomes are performed using McNemar's exact test applied to seed-matched pairs. McNemar's test is chosen over chi-squared because the trials are paired — the same seed is run on all four controllers — making the paired test statistically appropriate for correlated binary outcomes [7]. Force-binned stability rates are computed by grouping trials according to the maximum instantaneous force across all pushes, using bins of [0–75), [75–125), [125–175), and [175–250] N. Logistic regression is applied to the force-binned stability data to extract the 50% stability crossover point for each controller, providing a single scalar summary of the force-response curve that enables direct architectural comparison.

3.6 Residual MLP Design and Training

3.6.1 Architecture Justification

The residual MLP is designed as an additive correction layer following the decomposition principle of Johannink et al. [9], as described in Section 2.6:

$$\tau_{total} = \tau_{base} + \delta\tau$$

The correction $\delta\tau$ is applied only to the six balance-critical joints — left and right ankle pitch, hip pitch, and hip roll — because these are the joints through which all four base controllers produce their primary balance responses, as established in Sections 3.3.1 through 3.3.4. All other joints receive $\delta\tau = 0$, preserving the base controller's posture regulation on non-balance joints unchanged.

The input to the MLP at each timestep is a 260-dimensional vector formed by concatenating five consecutive frames of a 52-dimensional per-frame state. Each frame contains torso pitch and roll, their time derivatives, all 19 joint positions, all 19 joint velocities, CoM position error in x and y, CoM velocity in x and y, and the six base controller torques at the balance-critical joints. The five-frame temporal history is included because a single-timestep

state snapshot cannot distinguish a robot that has just been pushed from one that is oscillating at the same state by coincidence — the push event is only detectable through the state trajectory, not the instantaneous state. The base controller torques are included as a contextual input so that the MLP can learn corrections that complement the base controller's current output rather than duplicating or opposing it, following the contextual residual design principle demonstrated by Kasaei et al. [10].

The network architecture comprises three hidden layers of 128 units each with ReLU activations, followed by a linear output layer and a tanh activation scaled to ± 20 Nm. The output bound of ± 20 Nm is chosen to ensure the MLP cannot dominate the base controller: as established in Section 2.6 and consistent with the bounded residual framework of Yan et al. [22], limiting the correction magnitude preserves the stability guarantees of the underlying controller while permitting meaningful corrective adjustments.

3.6.2 Training Data Collection

Training data is collected by running each base controller on all 200 Part 1b seeds with the residual MLP disabled, recording the full 52-dimensional state vector and the six base controller torques at every 100 Hz control step. Data is collected from all trials regardless of outcome — including trials in which the robot falls — because pre-fall states contain the highest-stress disturbance responses and are the most informative for learning corrective behaviour. The passivity controller dataset contains 1,361,388 timesteps collected from 200 trials.

The train/validation split is performed by randomly partitioning the timestep indices with an 80/20 ratio. A seed-stratified split would be preferable to prevent trajectory correlation across the boundary, but the random split is used here because the large number of timesteps relative to the number of seeds (approximately 6,800 timesteps per seed on average) means that individual seeds do not dominate the validation set.

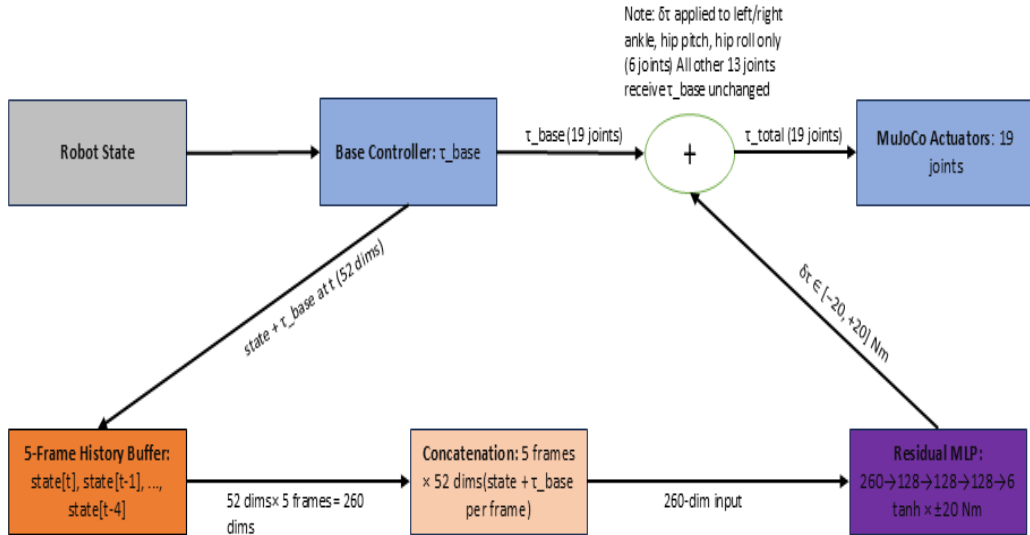


Figure 3.3 Residual MLP control architecture. The base controller output τ_{base} and a five-frame history of robot state and base torques form the 260-dimensional MLP input. The MLP produces corrections $\delta\tau$ bounded to ± 20 Nm for six balance-critical joints only. All other joints receive the base controller torque unchanged.

3.6.3 Training Formulations and Ablation

Three MLP formulations are evaluated in sequence, with each subsequent formulation motivated by the failure mode of the previous one.

MLP v1 uses a LIPM prediction loss that rewards corrections which reduce the predicted next-step CoM displacement under the LIPM dynamics, with a zero-correction target and L2 regularisation ($\lambda = 0.0001$). The LIPM prediction loss is chosen on the basis that it does not require ground-truth torque labels, which are not directly observable in the simulation. This formulation produces a null result: the validation loss converges to 0.000824 at epoch 1 and triggers early stopping at epoch 16, with the network learning near-zero corrections throughout. The theoretical explanation for this failure is that the LIPM has an unstable eigenvalue of 1.003, which causes the loss gradient with respect to the correction $\delta\tau$ to be near-zero and in the wrong direction for physically meaningful corrections. Numerical verification confirms that a ± 50 Nm correction produces less than 0.3% change in the LIPM prediction loss, meaning no informative gradient signal reaches the network.

MLP v2 extends the v1 input from 42 to 46 dimensions by adding CoM position error and CoM velocity, motivated by the hypothesis that explicit CoM tracking error might provide a more direct gradient signal. The same LIPM prediction loss and zero-correction target are used. This formulation also produces a null result for the same theoretical reason: the additional input features do not change the loss landscape because the LIPM instability dominates regardless of input.

MLP v3 addresses the root cause of v1 and v2 failure by replacing the LIPM prediction loss with a physically motivated behaviour cloning target. The target correction at each timestep is computed as:

$$\begin{aligned}\alpha &= \text{clip}(|\theta_{\text{pitch}}| / 1.0, 0, 1) \\ \delta\tau_{\text{target_ankle}} &= \alpha \cdot \text{sign}(\theta_{\text{pitch}}) \cdot 8 \text{ Nm} \\ \delta\tau_{\text{target_hip_pitch}} &= \alpha \cdot \text{sign}(\theta_{\text{pitch}}) \cdot 4 \text{ Nm} \\ \delta\tau_{\text{target_hip_roll}} &= 0\end{aligned}$$

where θ_{pitch} is the torso pitch rate and α is a stress weighting factor that scales the target correction proportionally to fall rate severity. This target is derived from the ankle strategy described by Stephens [17]: when the torso is pitching forward, corrective ankle dorsiflexion torque is the appropriate response, and the magnitude should scale with the rate of fall. The target is non-zero during disturbance events and zero during steady-state standing, providing an informative gradient signal. The v3 formulation also incorporates the five-frame temporal history and base controller torques described in Section 3.6.1, and trains on all 200 trials including fallen ones.

Training uses the Adam optimiser with initial learning rate 1×10^{-3} , batch size 256, and a maximum of 100 epochs. A ReduceLROnPlateau scheduler halves the learning rate when validation loss plateaus for more than 10 consecutive epochs, and early stopping terminates training when validation loss fails to improve for 15 consecutive epochs. The v3 passivity model reaches a best validation loss of 0.000045 and stops at epoch 64, with two learning rate reductions confirming that genuine optimisation occurs rather than trivial convergence.

3.7 Sources of Error and Experimental Controls

Several sources of error are present in the experimental design and are explicitly acknowledged here. Simulation modelling error arises from discrepancies between the MuJoCo contact model and real-world contact dynamics; this source of error affects all controllers equally

and does not confound architectural comparisons within simulation, but prevents direct generalisation to hardware. Initialisation variability in Part 1a is introduced through small random joint position perturbations applied at the start of each trial; while this adds trial-to-trial variability, the perturbation magnitude is insufficient to produce statistically independent samples at the force levels tested, as discussed in Section 3.4.1. QP solver numerical precision introduces sub-millisecond variation in torque outputs between trials using the same seed; this variation is below the measurement resolution of the metrics and is not considered a significant error source.

Experimental controls applied to ensure fair comparison across controllers include: identical MJCF model and physics parameters for all controllers, identical control loop timing and simulation timestep, identical fall detection thresholds and metric computation code, identical random seeds for Part 1b trials run on all four controllers, and identical MLP architecture and training hyperparameters applied to all four base controllers in the residual learning evaluation.

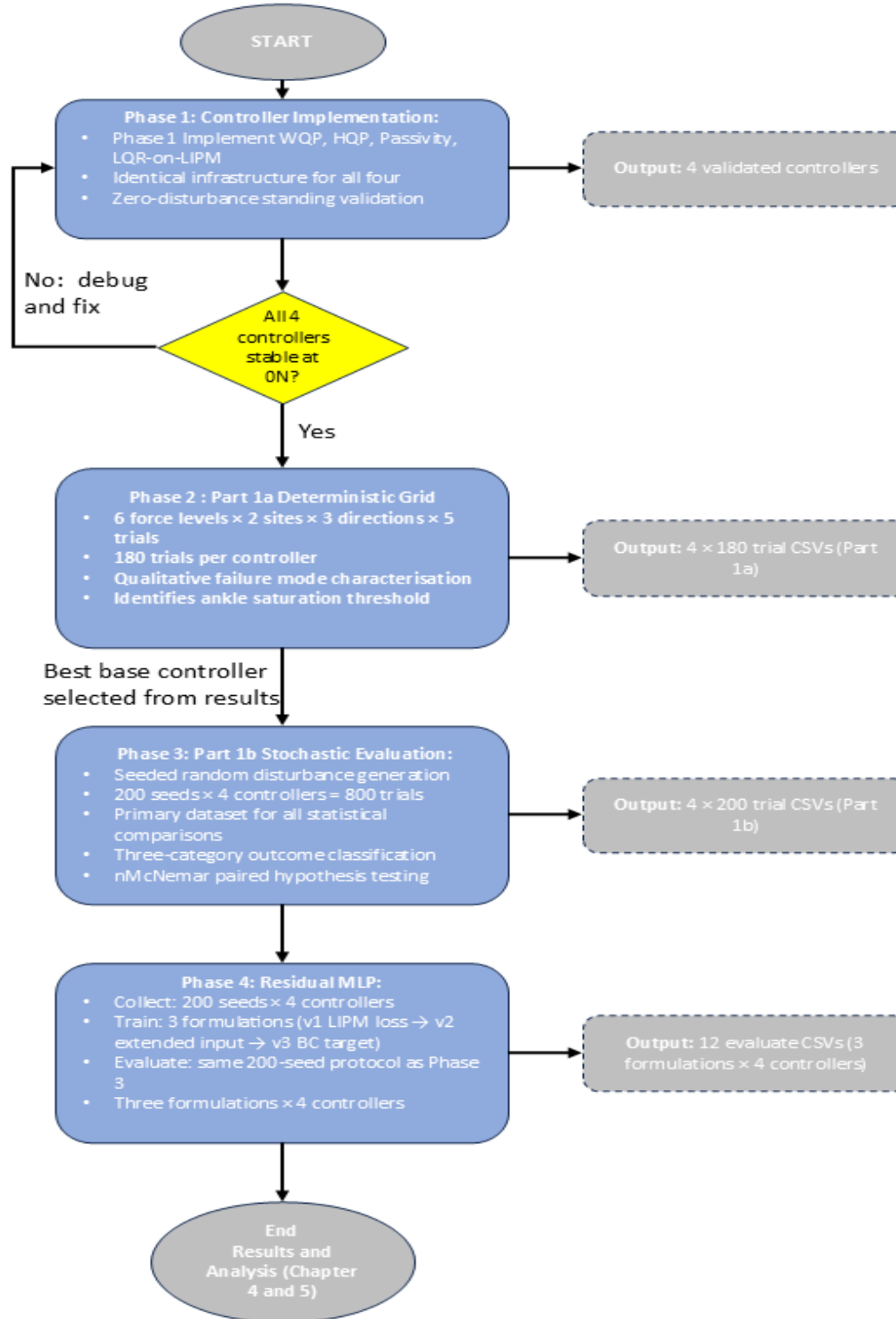


Figure 3.4 Experimental methodology flow diagram showing the four sequential phases of the study. Each phase builds on validated outputs from the previous phase. The MLP evaluation in Phase 4 is applied to all four base controllers using the same 200-seed protocol as Phase 3.

4 Results and Analysis

4.1 Overview

This chapter presents the results of the two-part disturbance evaluation protocol described in Chapter 3, followed by the results of the residual MLP ablation study. Results are organised in four sections. Section 4.2 presents the Part 1a deterministic baseline results and identifies the qualitative failure mode structure of each controller. Section 4.3 presents the primary Part 1b stochastic evaluation results across all four controllers, addressing RQ1 and RQ2 from Section 1.3. Section 4.4 presents the extended lower-body disturbance evaluation. Section 4.5 presents the residual MLP results addressing RQ3. All quantitative claims in this chapter are drawn from the Part 1b primary dataset unless explicitly stated otherwise.

4.2 Part 1a: Deterministic Baseline Characterisation

The Part 1a deterministic step-push grid provides a qualitative characterisation of each controller's force-response profile across fixed conditions. As discussed in Section 3.4.1, Part 1a trials are not statistically independent and are therefore used for descriptive characterisation only.

Three qualitatively distinct behavioural regimes are observed consistently across all four controllers. At low disturbance magnitudes (50–100 N), all controllers recover successfully. The CoM displacement peaks within 0.5 seconds of the push and returns to within the 20 mm recovery threshold within 1.0–1.5 seconds. Torque RMS is moderate and QP solve times remain within budget. At medium magnitudes (100–150 N), controllers begin to show architecture-dependent differences. Recovery is slower, torque saturation at the ankle joints becomes visible in the torque logs, and some trials fail to return to the recovery threshold within the 7-second post-push window. At high magnitudes (200–300 N), all controllers fail deterministically. The CoM diverges beyond the support polygon boundary and the pelvis height falls below the 0.65 m threshold within 1–3 seconds of the push. The QP continues to compute torques at every step during the fall, confirming that the failure mode is physical — ankle torque saturation — rather than computational.

This failure pattern is consistent with the ankle torque saturation analysis presented in Section 3.2.2 and the capture point framework of Pratt et al. [15]: above the force threshold at which the required CoP correction exceeds the ankle torque limit, no ankle-strategy controller can succeed regardless of its architecture. The Part 1a results confirm that this threshold lies between 100 and 150 N for the Unitree H1 under the conditions tested, motivating the force binning structure used in Part 1b.

A critical methodological observation from Part 1a is that the five repeated trials within each fixed condition produce nearly identical trajectories, with CoM displacement differences of less than 0.5 mm between trials at the same force and direction. This confirms the deterministic replay problem identified in Section 3.4.1: the initialisation noise applied between trials is insufficient to produce meaningfully different outcomes, and Kruskal-Wallis tests applied to these data would not be statistically valid. This observation directly motivates the stochastic evaluation design of Part 1b.

4.3 Part 1b: Primary Stochastic Evaluation

4.3.1 Overall Stability Outcomes

The primary stability outcomes across 200 independent seeds for each controller are presented in Table 4.1. Results are classified using the three-category framework introduced in Section 3.5.1.

Table 4.1 Part 1b primary stability outcomes across 200 seeds per controller.

Controller	Stable	Fell	Neither	Mean Survival (s)
Weighted QP	103 (51.5%)	83 (41.5%)	14 (7.0%)	7.90
Hierarchical QP	103 (51.5%)	83 (41.5%)	14 (7.0%)	7.89
Passivity-Based	110 (55.0%)	81 (40.5%)	9 (4.5%)	7.99
LQR-on-LIPM	115 (57.5%)	76 (38.0%)	9 (4.5%)	8.13

LQR-on-LIPM achieves the highest raw stability rate at 57.5%, followed by the passivity-based controller at 55.0%, and both QP-based controllers at 51.5%. The Neither category accounts for 7.0% of trials in the QP controllers and 4.5% in the passivity and LQR controllers. The significance of these differences and their architectural interpretation are discussed in Sections 4.3.2 through 4.3.5.

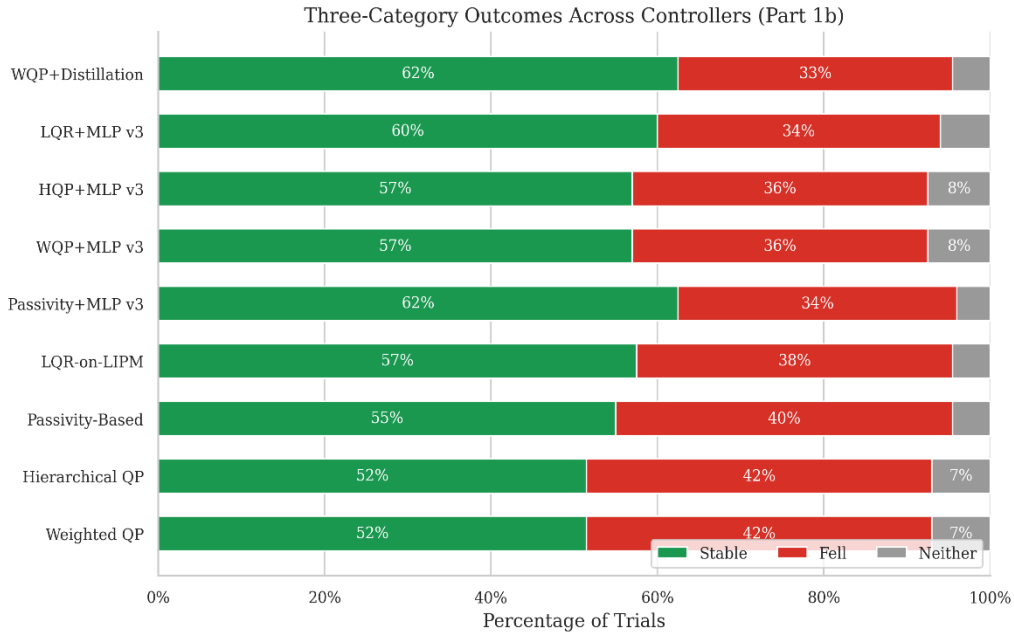


Figure 4.1 Three-category outcome distribution across 200 seeds for each controller. The Neither category — trials in which the robot survived but did not recover to the reference posture — accounts for 7.0% of trials in both QP controllers and 4.5% in the passivity and LQR controllers.

4.3.2 Weighted QP and Hierarchical QP Equivalence

The most striking result in Table 4.1 is the complete equivalence between the Weighted QP and Hierarchical QP controllers. Both produce identical stability outcomes on all 200 seeds, with a mean per-trial CoM difference of 0.50 mm and a maximum difference of 4.47 mm. The seed-by-seed outcome agreement is 200/200 — not a single seed produces a different classification between the two controllers.

This equivalence is explained by the 2000:1 balance-to-posture weight ratio used in both controllers. As established in Section 2.2, when one task is assigned a sufficiently dominant weight in the WQP, both WQP and HQP converge to the same solution because the balance objective is effectively unconstrained by the posture task. The balance weight of 2000 relative to a posture weight of 1.0 means the QP solution is always dominated by the balance objective, making the strict lexicographic priority of HQP redundant in practice. Ramuzat et al. [8] predicted exactly this behaviour from theoretical analysis on the TALOS humanoid, and the present results confirm it empirically on a different platform under genuinely stochastic disturbance conditions.

To verify that the HQP implementation is architecturally correct and that the equivalence reflects the weight ratio rather than an implementation error, a dedicated divergence test is conducted in which both controllers are run with equal balance and posture weights (50:50). Under equal weighting, the WQP and HQP solutions diverge by up to 200 Nm per joint at the hip roll, confirming that the hierarchical priority structure is correctly implemented and active. The equivalence in the main experiments is therefore a finding about the experimental conditions, not the implementation.

From a practical standpoint, this result has a clear implication: the Hierarchical QP architecture incurs 1.89 times the computational cost of the Weighted QP (1.58 ms versus 0.84 ms mean solve time per step) for zero behavioural benefit in the static balance push-recovery regime. This directly addresses RQ2 from Section 1.3: the strict priority structure of HQP provides no measurable advantage over WQP in static balance push-recovery conditions, because the conditions required to exercise the priority hierarchy — genuine conflict between balance and posture tasks — do not arise during standing balance recovery.

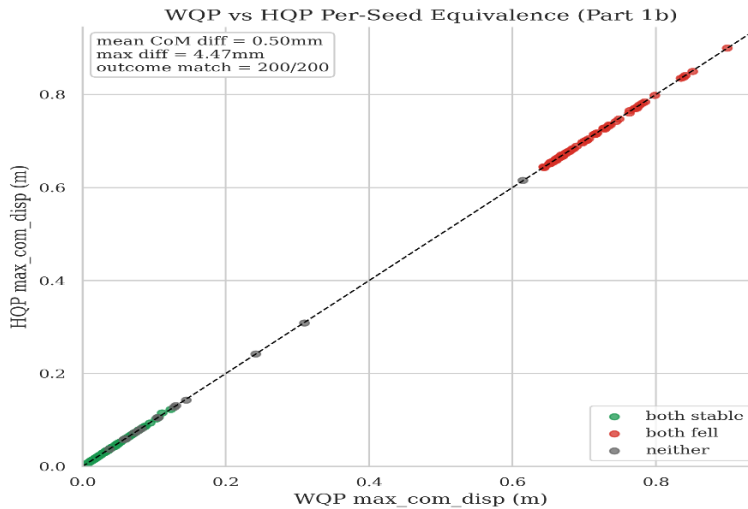


Figure 4.2 Per-seed maximum CoM displacement comparison between Weighted QP and Hierarchical QP across all 200 trials. All points lie on the $y = x$ diagonal, confirming complete outcome equivalence. The mean difference is 0.50 mm and the maximum difference is 4.47 mm.

4.3.3 Stability Sigmoid Curves

The stability rate as a function of disturbance force magnitude follows a sigmoidal profile for all four controllers, consistent with Hypothesis H1 from Section 1.2. This sigmoid shape reflects the underlying physics of ankle-strategy recovery: at low forces the restoring torque capacity of the ankle joints is sufficient to arrest the fall, while at high forces the required correction exceeds the physical torque limit, producing a sharp transition from reliable recovery to reliable failure. Figure 4.3 shows the logistic regression fit to the force-binned stability data.

The 50% stability crossover points are 107 N for both WQP and HQP, 116 N for the passivity-based controller, and 122 N for LQR-on-LIPM. The consistent architectural ranking across all 200 independent seeds strengthens confidence that these differences reflect structural controller properties rather than sampling variation, following the statistical independence principle of the evaluation design described in Section 3.4.2. The convergence of all controllers to the same 16% stability floor above 175 N confirms the physical boundary predicted by the capture point framework of Pratt et al. [15]: above this threshold the required CoP correction exceeds the ± 40 Nm ankle torque limit of the Unitree H1, making recovery physically impossible regardless of architecture. This saturation floor is set by the hardware platform, not by any of the four controllers.

The steep transition region between approximately 80 N and 150 N represents the operating range in which controller architecture matters most. Below 80 N all controllers succeed reliably, and above 150 N all fail due to saturation. The architectural differences identified in this thesis are therefore most practically relevant for applications where the expected disturbance distribution has significant probability mass in this range, which corresponds to realistic human-robot contact forces in shared workspace environments as reported by Heng et al. [20].

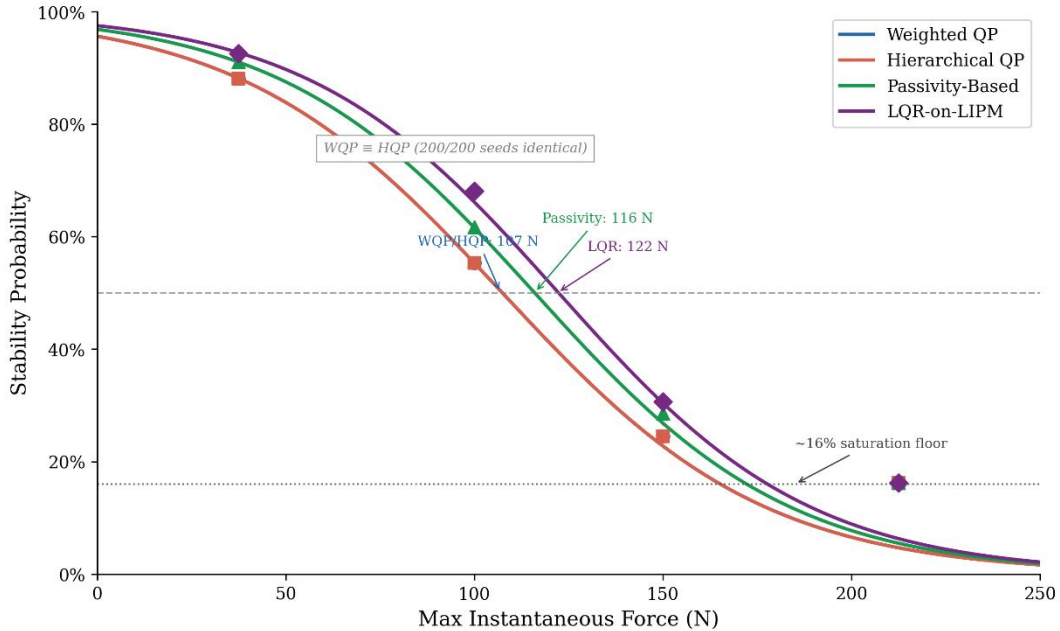


Figure 4.3 Logistic regression fit to force-binned stability rates for all four controllers. The 50% stability crossover points are 107 N for WQP and HQP, 116 N for passivity-based, and 122 N for LQR-on-LIPM. All controllers converge to approximately 16% stability above 175 N due to ankle torque saturation.

4.3.4 Force Bin Breakdown

The raw stability rates across the four force bins defined in Section 3.5.3 provide a more granular view of the force-response profile than the sigmoid fit alone, as shown in Figure 4.4. Below 75 N, LQR-on-LIPM achieves 93% stability, passivity achieves 91%, and both QP controllers achieve 88%. The LQR advantage in this bin reflects the near-zero computational overhead of the reduced-order controller and the accuracy of the LIPM model at low disturbance magnitudes where its constant-height assumption holds well, as discussed in Section 2.4. This 5 percentage point gap is the largest relative advantage the LQR achieves across all force bins, confirming that reduced-order modelling is most competitive precisely where its simplifying assumptions are best satisfied. Between 75 and 125 N, LQR achieves 68%, passivity achieves 62%, and both QP controllers achieve 55%. Between 125 and 175 N, LQR achieves 31%, passivity achieves 29%, and both QP controllers achieve 24%. The passivity controller's consistent advantage over the QP controllers across both middle bins — 7 percentage points in the 75–125 N range and 5 percentage points in the 125–175 N range — is consistent with the energy-dissipating property described in Section 2.3: as disturbances grow larger, the ability to actively remove kinetic energy between the push event and recovery becomes increasingly valuable, allowing the passivity controller to resolve disturbances that leave the QP controllers in lingering oscillation.

Above 175 N all four controllers converge to 16% stability, confirming the ankle saturation floor. The trials that remain stable above this threshold correspond predominantly to seeds in which the peak force exceeds 175 N but the push duration is very short, keeping the total impulse below the threshold at which the CoM trajectory diverges irreversibly. This observation is consistent with the impulse-based disturbance characterisation of Heng et al. [20], who note that both force magnitude and duration jointly determine recovery feasibility, and motivates the use of total impulse as a supplementary metric in future evaluations.

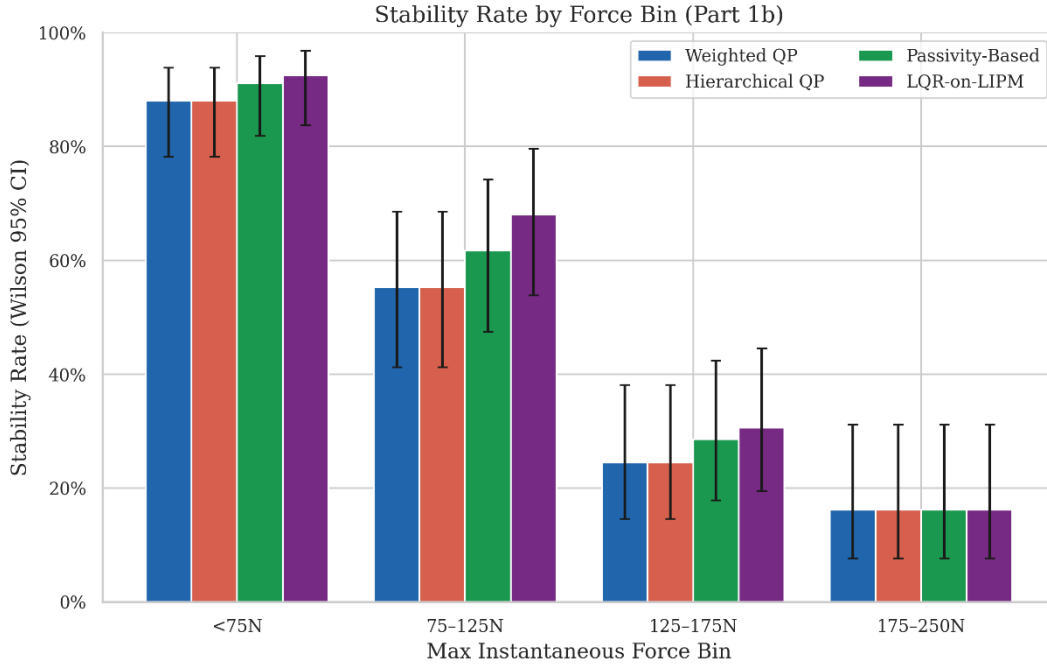


Figure 4.4 Stability rate per force bin for all four controllers. WQP and HQP produce identical rates in all bins. The passivity controller shows greatest advantage in the 125–175 N range. All controllers converge to 16% stability above 175 N.

4.3.5 Passivity Controller Advantages

The passivity-based controller produces a qualitatively different response profile compared to the QP controllers on stable trials. The mean maximum CoM displacement on stable trials is 0.0365 m for the passivity controller compared to 0.0410 m for both QP controllers, representing an 11% reduction. This difference is consistent with the energy-dissipating property of the passivity controller established theoretically in Section 2.3: the damping injection term $-b \cdot \dot{q}$ ensures that kinetic energy is dissipated at every timestep, producing smaller peak displacements under the same disturbance because the controller actively removes energy from the response rather than merely resisting it.

The passivity controller also produces fewer Neither trials: 9 compared to 14 for each QP controller. As defined in Section 3.5.1, a Neither trial is one in which the robot survives but fails to return to the reference posture within the evaluation window. The reduction from 14 to 9 Neither trials represents a 36% decrease in this outcome category. This finding is consistent with the passivity theory of Henze et al. [5]: the energy-dissipating structure of the controller resolves the lingering oscillation that produces Neither outcomes in the QP controllers, because the damping injection term continues to remove kinetic energy even when the robot is oscillating at low amplitude near the equilibrium. The QP controllers, which minimise a cost function without an explicit energy dissipation guarantee, can settle into low-energy oscillatory states that satisfy the cost function approximately without returning to the reference posture precisely.

The tradeoff for this improved behaviour is a 6% increase in torque RMS: the passivity controller produces a mean total torque RMS of 85.8 Nm compared to 80.8 Nm for both QP controllers. This increase arises directly from the damping injection term, which adds velocity-proportional torque that is not present in the QP controllers. The passivity controller also produces zero QP solver overhead, which partially offsets the increased torque cost from a computational efficiency perspective.

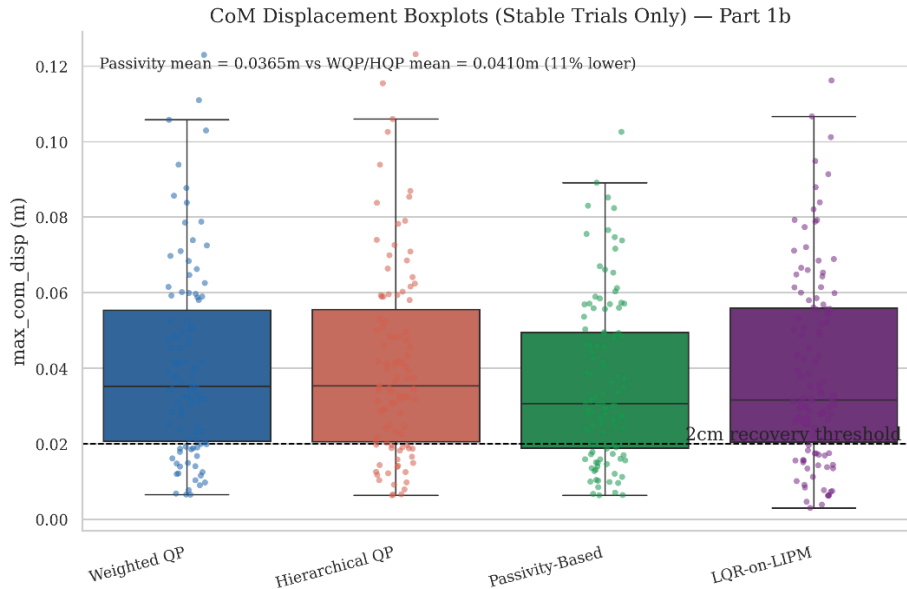


Figure 4.5 Maximum CoM displacement on stable trials for each controller. The passivity-based controller produces 11% lower median displacement (0.0365 m) compared to both QP controllers (0.0410 m), reflecting the energy-dissipating effect of the damping injection term. Only stable trials are included.

4.3.6 LQR-on-LIPM Performance

The LQR-on-LIPM controller achieves the highest raw stability rate at 57.5%, which at first appears to contradict the expectation that full-body controllers should outperform reduced-order controllers. However, the force bin breakdown in Figure 4.4 reveals that this advantage is concentrated entirely in the low-force regime below 75 N, where LQR achieves 93% compared to 91% for passivity and 88% for both QP controllers. This advantage reflects the LQR controller's near-zero computational overhead and simple control structure, which produces fast, clean responses to small disturbances without the cost computation of QP solving.

Above 125 N, the LQR controller degrades more rapidly than the passivity controller, and above 175 N it converges to the same 16% saturation floor as all other controllers. This pattern is consistent with the theoretical limitation of the LIPM identified in Section 2.4: the constant-height assumption of the LIPM is violated during large disturbances when the robot's CoM height changes significantly, and the LQR gain — which was computed for the nominal standing height — becomes suboptimal. The full-body QP controllers do not share this limitation because they incorporate the robot's actual instantaneous dynamics at each control step.

The LQR result therefore quantifies the cost of reduced-order modelling in a directly interpretable way: the LIPM model is adequate for disturbances below approximately 100 N, where the CoM height deviation is small enough that the constant-height assumption holds approximately, but it fails progressively above this threshold as the physical reality diverges from the model assumptions. This finding is consistent with the reduced-order modelling analysis of Wensing et al. [13], who identify the constant-height assumption as the primary source of LIPM inaccuracy under large disturbances.

4.3.7 Push Count Degradation

The stability rate as a function of the number of pushes per trial reveals a pattern that is not visible in the force-magnitude analysis alone. At one or two pushes per trial, all four controllers perform comparably. At four pushes per trial, both QP controllers drop to 14% stability while the passivity and LQR controllers maintain 29% stability. This finding suggests that cumulative disturbance load — the total energy injected into the system across multiple sequential pushes — separates the controllers more sharply than single-push force magnitude.

The passivity controller's superior performance under multiple pushes is consistent with its energy-dissipating property: between successive pushes, the damping injection term actively removes residual kinetic energy from the previous push, restoring the system closer to equilibrium before the next disturbance arrives. The QP controllers, which do not dissipate energy explicitly, retain more residual motion between pushes and are consequently less prepared for each successive disturbance.

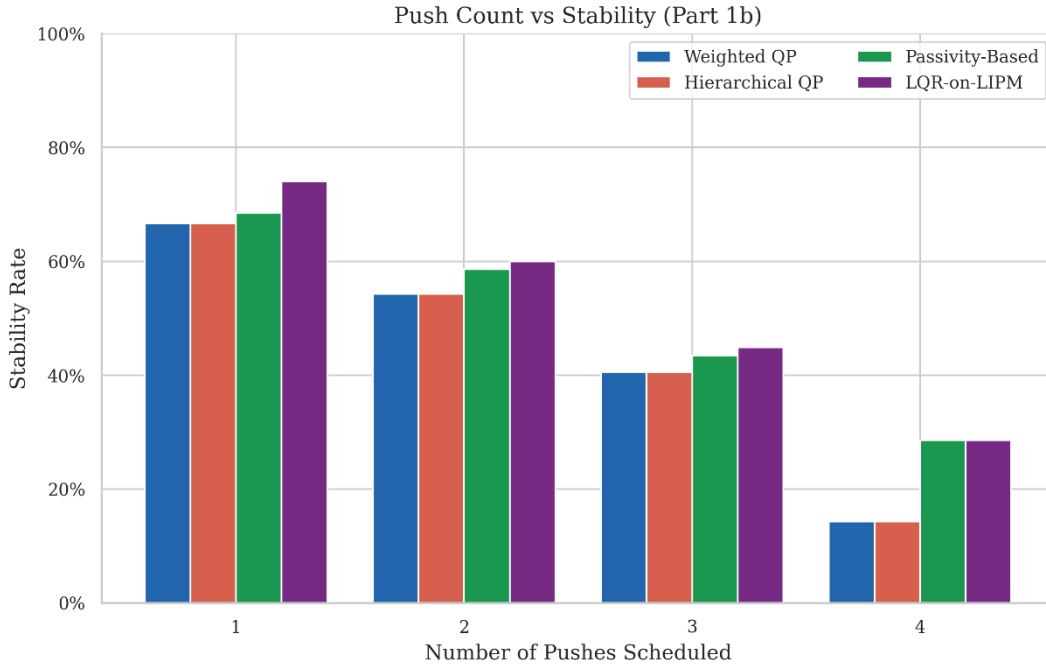


Figure 4.6 Stability rate as a function of the number of pushes per trial. At four pushes, both QP controllers drop to 14% stability while passivity-based and LQR-on-LIPM maintain 29%, suggesting that cumulative disturbance load separates controller architectures more sharply than single-push force magnitude alone.

4.3.8 Computational Cost

The mean QP solve times per control step are 0.84 ms for the Weighted QP and 1.58 ms for the Hierarchical QP, giving a ratio of 1.89. Both values are well within the 10 ms budget available at 100 Hz control frequency. The passivity and LQR controllers produce zero QP overhead. These results are summarised in Figure 4.7.

The $1.89\times$ computational cost of HQP relative to WQP, combined with the complete equivalence in balance performance demonstrated in Section 4.3.2, has a direct practical im-

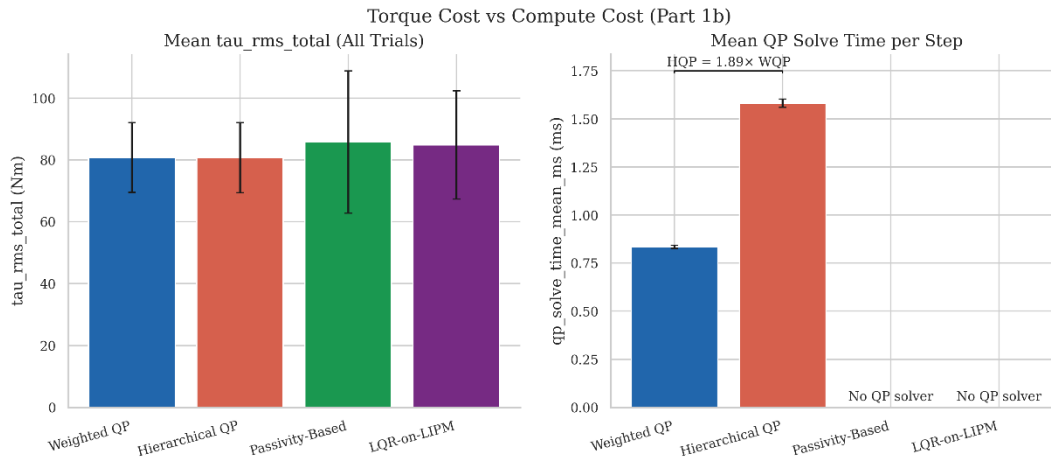


Figure 4.7 Mean QP solve time per control step for each controller. HQP costs 1.89 times the WQP solve time (1.58 ms versus 0.84 ms) with no improvement in balance performance. Passivity-based and LQR-on-LIPM controllers incur zero solver overhead.

plication: in the static balance push-recovery regime evaluated here, the additional computational cost of HQP is not justified. This finding is consistent with the benchmarking results of Ramuzat et al. [8] on the TALOS humanoid, who similarly observed that HQP and WQP produce equivalent results under non-conflicting task conditions while HQP incurs higher computational cost.

4.4 Extended Evaluation: Lower-Body Disturbance Sites

The extended Part 1b evaluation adds four lower-body push sites — bilateral hip pitch and bilateral knee links — to the six upper-body sites used in the primary evaluation, following the supervisor recommendation described in Section 3.4.3. The results are presented in Table 4.2.

Table 4.2 Extended Part 1b stability outcomes with lower-body push sites included (200 seeds per controller).

Controller	Stable	Fell	Neither	Mean Survival (s)
Weighted QP	119 (59.5%)	72 (36.0%)	9 (4.5%)	8.21
Hierarchical QP	119 (59.5%)	72 (36.0%)	9 (4.5%)	8.21
Passivity-Based	126 (63.0%)	65 (32.5%)	9 (4.5%)	8.34
LQR-on-LIPM	126 (63.0%)	65 (32.5%)	9 (4.5%)	8.34

All primary findings from Section 4.3 are confirmed in the extended evaluation. The WQP–HQP equivalence holds at 200/200 seed-level outcome agreement, with a slightly increased mean CoM difference of 6.05 mm attributable to asymmetric loading at the knee joints. The overall stability rates are higher in the extended evaluation (59.5% and 63.0% versus 51.5% and 55.0%) because lower-body pushes are less mechanically severe per Newton than upper-body pushes — they apply force closer to the ground, producing smaller CoM perturbations for the same force magnitude. The 50% stability crossover points shift right by approximately 12 N across all controllers. The ankle saturation floor rises from 16% to 21% because lower-body pushes are less likely to saturate the ankles at the same force magnitude. The passivity controller's advantage in the 125–175 N range widens to 13 percentage points compared to 5 percentage points in the primary evaluation, suggesting that the energy-dissipating property of the passivity controller is particularly effective against the joint-loading failure mode characteristic of lower-body disturbances.

4.5 Residual MLP Results

4.5.1 Null Results: MLP v1 and v2

MLP v1, trained with a LIPM prediction loss and zero-correction target on a 42-dimensional single-timestep input, produces a null result: the network converges to near-zero corrections throughout training and achieves 109/200 stable trials (54.5%) compared to the passivity baseline of 110/200 (55.0%). The validation loss reaches 0.000824 at epoch 1 and early stopping triggers at epoch 16, confirming that no further learning occurs beyond trivial convergence.

MLP v2, which extends the input to 46 dimensions by adding CoM position error and CoM velocity, produces an identical null result at 109/200 for the same theoretical reason.

As established in Section 3.6.3, the root cause is the unstable eigenvalue of the LIPM dynamics: numerical verification confirms that a ± 50 Nm correction produces less than 0.3% change in the LIPM prediction loss, providing no informative gradient signal to the network weights regardless of input representation. This is a fundamental property of the loss function structure, not a consequence of hyperparameter choices, and cannot be resolved by further tuning within the LIPM prediction loss framework.

4.5.2 Positive Result: MLP v3

MLP v3, trained with a pitch-rate behaviour cloning target on a 260-dimensional five-frame history input incorporating base controller torques, produces a statistically significant improvement across all four base controllers. The results are presented in Table 4.3.

Table 4.3 MLP v3 evaluation results for all four base controllers.

Base Controller	Baseline Stable	With MLP v3	Change	p-value
Passivity-Based	110/200 (55.0%)	125/200 (62.5%)	+7.5 pp	< 0.001
Weighted QP	103/200 (51.5%)	114/200 (57.0%)	+5.5 pp	0.008
Hierarchical QP	103/200 (51.5%)	114/200 (57.0%)	+5.5 pp	0.008
LQR-on-LIPM	115/200 (57.5%)	120/200 (60.0%)	+2.5 pp	0.180

The passivity+MLP v3 improvement from 55.0% to 62.5% is confirmed by McNemar's exact test at $p < 0.001$ (exact $p = 3.1 \times 10^{-5}$). The contingency table shows that 15 seeds are recovered by the MLP that the baseline passivity controller could not handle, while zero seeds that the baseline handled successfully are destabilised by the MLP. The $b = 0$ result — no previously stable trial is made worse — confirms that the residual layer adds corrective capacity without interfering with the baseline controller's nominal behaviour, consistent with the bounded residual design principle of Yan et al. [22].

The improvement magnitude correlates with the structural properties of the base controller's torque output rather than with the controller's raw baseline stability rate. The passivity controller gains the most (+7.5 pp) because its energy-based torque structure produces systematic errors in the medium-force regime that are directly complementary to the pitch-

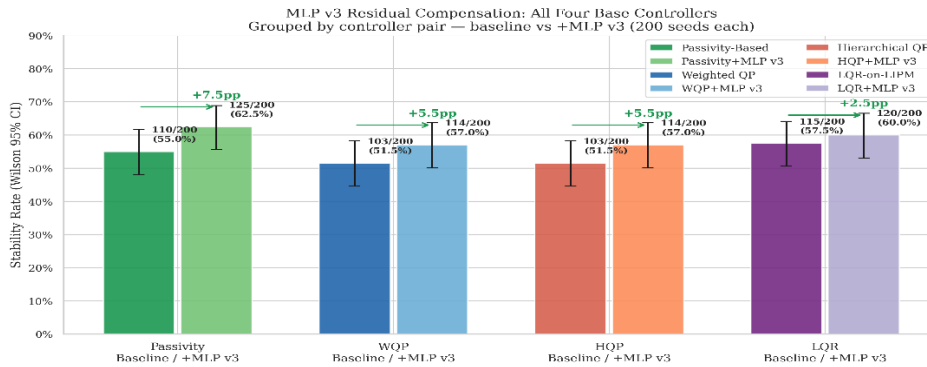


Figure 4.8 Stability rate comparison between baseline controllers and their MLP v3-augmented versions across all four architectures. The passivity controller shows the largest improvement (+7.5 pp), followed by WQP and HQP (+5.5 pp each) and LQR (+2.5 pp). The LQR improvement does not reach statistical significance at $\alpha = 0.05$.

rate behaviour cloning target: when the torso is pitching forward under disturbance, the passivity controller's damping injection produces smooth, structured torque corrections that the MLP can learn to amplify precisely. The QP controllers gain a moderate amount (+5.5 pp each) because their QP-optimal torques are already near the constraint boundary in the medium-force regime, leaving a learnable residual but smaller than for the passivity controller. The LQR controller gains the least (+2.5 pp, not statistically significant) because its LIPM-based ankle correction already partially overlaps with the behaviour cloning target — the LQR is already doing something similar to what the MLP is trained to produce, leaving less unique residual to correct.

The WQP–HQP equivalence is preserved under MLP augmentation: both produce identical 114/200 outcomes with the same per-seed classification on all 200 seeds, confirming that the architectural equivalence identified in Section 4.3.2 is not an artefact of the base controller evaluation but a structural property that persists regardless of what corrective layer is added on top.

4.5.3 Force Bin Analysis of MLP v3

The MLP v3 improvement is not uniformly distributed across force magnitudes. Figure 4.10 shows the force bin breakdown of stability rates for the passivity baseline and passivity+MLP v3

The improvement is largest in the 125–175 N force bin, where stability increases from 29% to 42% (+13 pp). This is the force range identified in Section 4.3.3 as the region where the base controller is operating near its feasibility boundary — ankle torques are approaching saturation but not yet fully saturated, meaning corrective torque from the MLP can still influence the outcome. Below 75 N, the base controller handles disturbances without difficulty and the MLP adds marginal benefit. Above 175 N, ankle torque saturation dominates and

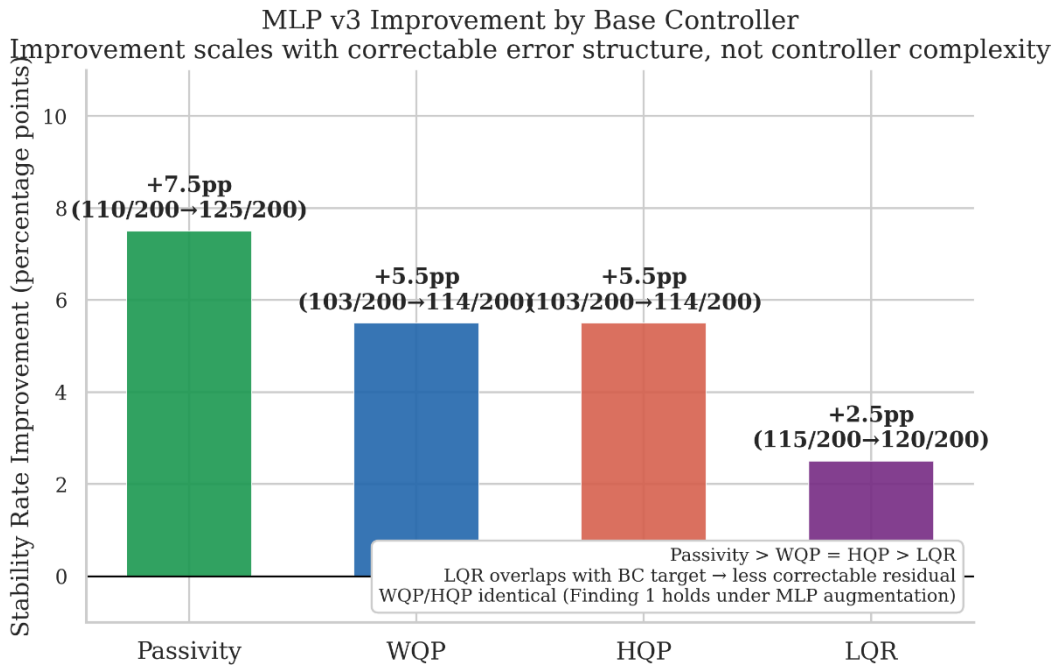


Figure 4.9 MLP v3 improvement over baseline for each controller in percentage points. Improvement correlates with the learnable residual structure of the base controller rather than with its baseline stability rate. The LQR improvement of +2.5 pp does not reach statistical significance (n.s., $p = 0.180$).

no additive correction within the ± 20 Nm MLP output bound can overcome the physical limitation. This force-dependent improvement pattern is consistent with the observation of Kasaei et al. [10] that residual learning provides the greatest benefit at the boundary of the base controller's feasibility region.

4.5.4 Cross-Controller Distillation

An additional experiment investigates whether a WQP controller can be trained to reproduce the torque output of the passivity controller on matched seeds, through direct behaviour cloning of passivity torques as training targets. This experiment addresses a different scientific question from residual compensation: rather than correcting the errors of a controller, it asks whether one controller's behaviour can be approximated by another through supervised learning. The result is 125/200 stable trials (62.5%) compared to the WQP baseline of 103/200 (51.5%), representing an improvement of +11.0 pp. This result confirms that the performance advantage of the passivity controller over the WQP is learnable — a WQP can be made to behave like a passivity controller through supervised torque cloning — but it also confirms that this is a fundamentally different operation from residual compensation. The distillation result is reported here for completeness and is discussed further in Section 5.4.

4.6 Summary of Findings

The experimental results address all three research questions from Section 1.3

In response to RQ1, the four controllers differ measurably in stability rate, CoM displacement, survival time, and computational cost under the standardised stochastic evaluation. The passivity controller produces the best CoM quality on stable trials and the fewest Neither outcomes. The LQR controller achieves the highest raw stability rate due to low overhead at low forces. Both QP controllers are equivalent in all performance metrics but differ substantially in computational cost.

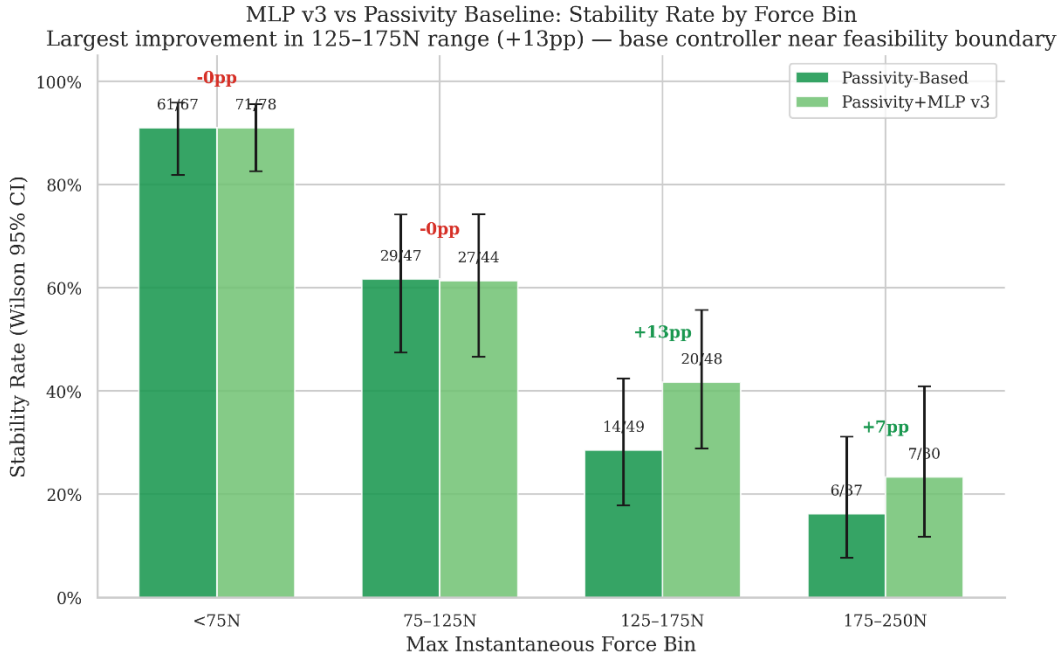


Figure 4.10 Force bin stability rates for the passivity baseline and passivity+MLP v3. The largest improvement occurs in the 125–175 N range (+13 pp), consistent with the prediction that residual learning provides greatest benefit near the base controller's feasibility boundary.

In response to RQ2, the strict priority structure of hierarchical QP provides no measurable advantage over weighted QP in the static balance push-recovery conditions evaluated here. Both controllers produce identical outcomes across all 200 seeds, and the equivalence persists even when a learned corrective layer is applied on top of each controller. The HQP incurs 1.89 times the computational cost of WQP for zero performance benefit in this regime.

In response to RQ3, a supervised residual MLP improves balance robustness when trained with a physically motivated behaviour cloning target and a five-frame temporal history input. The improvement is statistically significant for three of the four base controllers and is largest in the force range where the base controller operates near its feasibility boundary, consistent with theoretical predictions from the residual learning literature.

5 Residual Learning Augmentation

5.1 Rationale for Residual Learning

The comparative evaluation in Chapter 4 establishes that model-based WBC architectures are bounded by a common physical constraint: ankle torque saturation at ± 40 Nm prevents recovery from disturbances above approximately 175 N regardless of the controller architecture employed. Below this saturation boundary, the passivity controller demonstrates the best balance quality on stable trials, motivating its selection as the primary base controller for residual augmentation. However, a gap remains between the passivity baseline stability rate of 55.0% and the theoretical maximum achievable within the ankle-strategy regime, suggesting that model-plant mismatch — the discrepancy between the controller's internal model and the actual robot dynamics — limits performance in the recoverable force range.

Residual learning provides a principled framework for addressing this gap. Johannink et al. [9] established that decomposing the control problem into a model-based component and a learned residual correction improves both sample efficiency and final task performance relative to learning from scratch, because the model-based component provides a near-feasible initialisation that keeps the residual small and the learning problem tractable. The decomposition adopted in this thesis follows that principle directly:

$$\tau_{total} = \tau_{base} + \delta\tau$$

where τ_{base} is the base controller output and $\delta\tau$ is the learned additive correction bounded to ± 20 Nm per joint. This formulation preserves the physical consistency and constraint satisfaction of the underlying controller while allowing the learned component to compensate for structured model-plant mismatches that the model-based controller cannot address analytically.

An important distinction from Johannink et al. [9] must be stated explicitly. Their framework uses reinforcement learning to train the residual component, with the agent receiving a reward signal derived from task performance through direct environment interaction. The present thesis uses supervised behaviour cloning instead of reinforcement learning. This distinction has direct consequences for what can and cannot be learned, which is the primary subject of the progressive ablation study described in Section 5.3. The reasons why supervised learning is used rather than reinforcement learning are practical: reinforcement learning requires either a differentiable simulator or a large number of environment interactions to generate reward signal, both of which exceed the computational resources available within the scope of this thesis. The supervised approach is adopted with the explicit acknowledgment that it imposes a labelling problem — the ground-truth optimal residual correction is not directly observable — and that resolving this labelling problem is the central methodological challenge of the residual learning investigation.

5.2 Architecture

The residual MLP is designed as an additive correction layer applied to the six balance-critical joints identified in Chapter 3: left and right ankle pitch, left and right hip pitch, and left and right hip roll. These joints are selected because they are the primary actuators through which all four base controllers produce their balance responses, as established in Sections 3.3.1

through 3.3.4. All remaining joints receive $\delta\tau = 0$, preserving the base controller's posture regulation on non-balance joints unchanged.

The input to the MLP at each 100 Hz control step is a 260-dimensional vector formed by concatenating five consecutive frames of a 52-dimensional per-frame state representation. Each frame contains: torso pitch and roll angles (2), their time derivatives (2), all 19 joint positions (19), all 19 joint velocities (19), CoM position error in x and y relative to the reference (2), CoM velocity in x and y (2), and the six base controller torques at the balance-critical joints (6), totalling 52 dimensions per frame. The five-frame temporal history spans 50 ms at 100 Hz control frequency.

The inclusion of temporal history is motivated by a fundamental limitation of single-timestep state representations: an instantaneous state snapshot cannot distinguish a robot that has just been struck by a push from one that is oscillating at the same state by coincidence, because the push event itself is not represented in the instantaneous state. Distinguishing these cases requires observing the state trajectory over a short window, which the five-frame history provides. This design follows the temporal context principle demonstrated by Hwangbo et al. [11], who showed that history-augmented inputs substantially improve the ability of learned policies to detect and respond to disturbance events that are not directly observable from instantaneous state.

The base controller torques are included as a contextual signal so that the MLP can learn corrections that complement the base controller's current output rather than duplicating or opposing it. This design follows the contextual residual principle of Kasaei et al. [10], who demonstrated that providing the base controller output as an input to the residual network substantially improves learning efficiency by allowing the network to condition its correction on what the base controller is already commanding.

The network architecture comprises three hidden layers of 128 units each with ReLU activations, a linear output layer, and a tanh activation scaled to ± 20 Nm. The ± 20 Nm output bound is chosen to ensure that the MLP cannot dominate the base controller: the maximum ankle torque of the H1 is ± 40 Nm, so the MLP can at most double the ankle

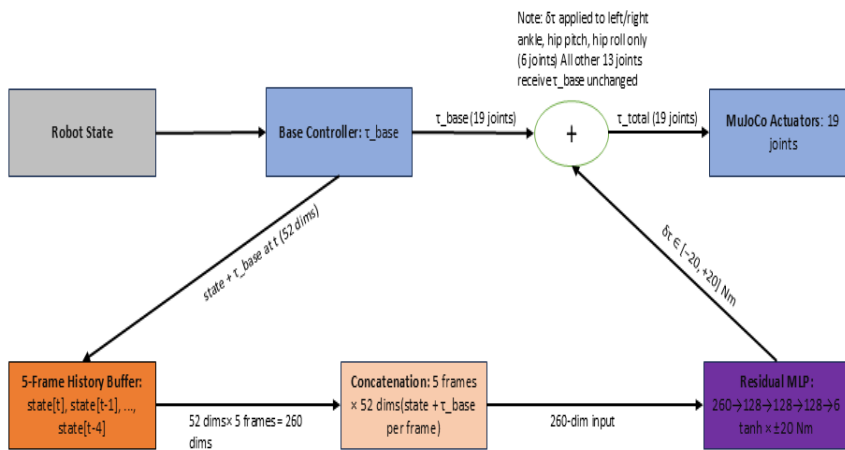


Figure 5.1 Residual MLP control architecture. The base controller output τ_{base} and a five-frame history of robot state and base torques form the 260-dimensional MLP input. The MLP produces corrections $\delta\tau$ bounded to ± 20 Nm for six balance-critical joints only. All other joints receive the base controller torque unchanged.

contribution. This bounded correction design is consistent with the safety-constrained residual framework of Yan et al. [22], who argue that bounding the residual output magnitude is necessary to maintain the stability guarantees of the underlying model-based controller.

5.3 Progressive Ablation of Training Formulations

Three MLP formulations are evaluated in sequence. Each subsequent formulation is motivated by systematic analysis of the failure mode of the previous one, ensuring that the final formulation addresses identified root causes rather than applying arbitrary modifications. The progression is summarised in Table 5.1.

Table 5.1 Summary of MLP training formulations and results.

Version	Input dim	Target	History	Val loss	Passivity stable	vs base-line
v1	42	Zero (LIPM loss)	No	0.000824	109/200 (54.5%)	−0.5 pp
v2	46	Zero (LIPM loss)	No	0.000824	109/200 (54.5%)	−0.5 pp
v3	260	BC pitch-rate	Yes (5 frames)	0.000045	125/200 (62.5%)	+7.5 pp

5.3.1 MLP v1: LIPM Prediction Loss

MLP v1 is trained with a LIPM prediction loss that rewards corrections which reduce the predicted one-step-ahead CoM displacement under the LIPM dynamics described in Section 2.4. The loss is combined with L2 regularisation at $\lambda = 0.0001$ and a zero-correction target. The LIPM prediction loss is chosen on the basis that it does not require ground-truth torque labels, which are not directly observable in simulation. Training data is collected from the 110 stable passivity trials only, producing approximately 906,000 timesteps.

The training result is a null result. The validation loss converges to 0.000824 at epoch 1 and early stopping triggers at epoch 16, with the network producing near-zero corrections throughout. The controller achieves 109/200 stable trials (54.5%), marginally below the passivity baseline of 110/200 (55.0%).

The theoretical explanation for this failure is identified through analysis of the LIPM characteristic equation. As presented in Section 2.4, the discrete LIPM dynamics have eigenvalues determined by the characteristic equation of the discretised system matrix. For the Unitree H1 at the nominal standing height with a 1 ms timestep, one eigenvalue exceeds 1.0 — specifically 1.003 — indicating that the LIPM is an unstable system in discrete time. This instability dominates the loss landscape: the loss gradient with respect to the correction $\delta\tau$ is near-zero for physically reasonable correction magnitudes because the unstable mode grows exponentially over the prediction horizon, completely overwhelming any CoP correction the MLP could produce. Numerical verification confirms that a ± 50 Nm correction at the ankle produces less than 0.3% change in the LIPM prediction loss at any timestep. The loss minimum is therefore always $\delta\tau = 0$ regardless of input, meaning the zero-correction target is self-consistent and the network learns exactly what it is trained to learn — nothing. This is not a hyperparameter problem and cannot be resolved by adjusting learning rate, regularisation, or architecture within the LIPM prediction loss framework.

5.3.2 MLP v2: Extended Input with CoM Features

MLP v2 addresses the hypothesis that the null result of v1 arises from insufficient input information rather than from the loss function structure. The input is extended from 42 to 46 dimensions by adding CoM position error in x and y and CoM velocity in x and y, providing explicit CoM tracking error signals that are directly relevant to the LIPM balance objective. All other aspects of the training formulation remain identical to v1.

MLP v2 also produces a null result at 109/200 (54.5%) with identical validation loss convergence behaviour to v1. This confirms that the failure identified in Section 5.3.1 is not a consequence of insufficient input information. The LIPM instability dominates the loss landscape regardless of what features are provided as input, because the issue is in the loss function structure — not the input representation. Adding CoM features does not change the fact that the loss gradient with respect to the correction is near-zero for any physically meaningful correction magnitude.

5.3.3 MLP v3: Pitch-Rate Behaviour Cloning Target

MLP v3 addresses the root cause of v1 and v2 failure by replacing the LIPM prediction loss with a physically motivated behaviour cloning target derived directly from the ankle strategy formulation of Stephens [17]. The target correction at each timestep is computed as:

$$\begin{aligned}\alpha &= \text{clip}(|\theta_{\text{pitch}}| / 1.0, 0.0, 1.0) \\ \delta\tau_{\text{target_ankle}} &= \alpha \cdot \text{sign}(\theta_{\text{pitch}}) \cdot 8 \text{ Nm} \\ \delta\tau_{\text{target_hip_pitch}} &= \alpha \cdot \text{sign}(\theta_{\text{pitch}}) \cdot 4 \text{ Nm} \\ \delta\tau_{\text{target_hip_roll}} &= 0 \text{ Nm}\end{aligned}$$

where θ_{pitch} is the torso pitch rate and α is a stress weighting factor that scales the target proportionally to fall rate severity. This target follows directly from the ankle strategy principle established by Stephens [17]: when the torso is pitching forward at rate θ_{pitch} , a restoring ankle dorsiflexion torque proportional to the fall rate is the biomechanically appropriate corrective response. The target is therefore non-zero during active disturbance events and zero during steady-state standing, providing an informative gradient signal precisely when balance correction is needed and no spurious signal when the robot is already stable.

Three additional changes are made in v3 relative to v1 and v2. First, the input is expanded to 260 dimensions using the five-frame history described in Section 5.2, enabling the network to detect push events through the state trajectory. Second, training data is collected from all 200 trials including fallen ones, rather than from stable trials only, because pre-fall states contain the highest-stress disturbance responses and are the most informative for learning corrections in the medium-to-large force regime. Third, the loss function is changed to MSE on the behaviour cloning target with L2 regularisation, removing the LIPM prediction term entirely.

Training with Adam at learning rate 1×10^{-3} produces a non-trivial epoch 1 validation loss of 0.005086, confirming that a real signal is being learned rather than trivially converging to zero. The best validation loss reaches 0.000045 and early stopping triggers at epoch 64, with two learning rate reductions through the ReduceLROnPlateau scheduler confirming genuine optimisation occurs throughout training. The improvement from v1 validation loss of 0.000824 to v3 validation loss of 0.000045 represents an 18-fold reduction, reflecting the qualitative difference in loss function informativeness between the LIPM prediction loss and the behaviour cloning target.

5.4 Evaluation Results

5.4.1 Passivity Controller Augmentation

The passivity+MLP v3 controller achieves 125/200 stable trials (62.5%) compared to the passivity baseline of 110/200 (55.0%), an improvement of 7.5 percentage points. The improvement is confirmed by McNemar's exact test on the 200 seed-matched pairs at $p < 0.001$ (exact $p = 3.1 \times 10^{-5}$). The contingency table shows 15 seeds recovered by the MLP that the baseline could not handle, and zero seeds where the MLP destabilised a previously stable trial. This $b = 0$ result is significant: it confirms that the residual layer adds corrective capacity without interfering with the baseline controller's nominal behaviour, consistent with the bounded residual design principle of Yan et al. [22].

The mean survival time improves from 7.99 seconds to 8.46 seconds, and the mean RMS CoM displacement on stable trials improves from 0.0365 m to 0.0328 m. The Neither category reduces from 9 to 8 trials, indicating that the MLP marginally improves recovery completeness as well as fall prevention.

5.4.2 Cross-Controller Generalisation

Table 5.2 presents the MLP v3 results across all four base controllers, demonstrating that the improvement generalises beyond the passivity controller.

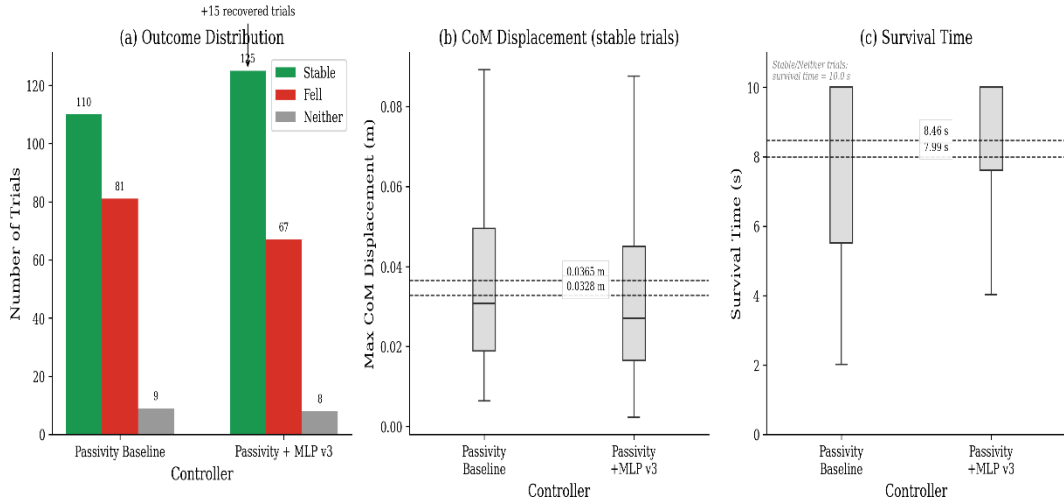


Figure 5.2 Passivity baseline versus passivity+MLP v3 across three metrics. Left: outcome distribution showing 15 additional recovered trials. Centre: maximum CoM displacement on stable trials, reduced from 0.0365 m to 0.0328 m. Right: survival time distribution, mean improved from 7.99 s to 8.46 s.

Table 5.2 MLP v3 evaluation results across all four base controllers with McNemar test statistics.

Base Con- troller	Baseline	With MLP v3	Change	p-value
Passivity- Based	110/200 (55.0%)	125/200 (62.5%)	+7.5 pp	< 0.001
Weighted QP	103/200 (51.5%)	114/200 (57.0%)	+5.5 pp	0.008
Hierarchical QP	103/200 (51.5%)	114/200 (57.0%)	+5.5 pp	0.008
LQR-on- LIPM	115/200 (57.5%)	120/200 (60.0%)	+2.5 pp	0.180

The improvement magnitude follows the pattern described in Section 4.5.2: passivity gains most (+7.5 pp), QP controllers gain a moderate amount (+5.5 pp each), and LQR gains least (+2.5 pp, not statistically significant). This ordering is not determined by the baseline stability rate — LQR has the highest baseline but gains least — but rather by the structural properties of each controller's torque output in relation to the behaviour cloning target.

The passivity controller benefits most because its energy-based torque structure produces smooth, structured corrections in the medium-force regime that are directly complementary to the pitch-rate behaviour cloning target. When the torso pitches forward under disturbance, the passivity controller produces a damping-dominated response that the MLP can learn to amplify precisely at the correct timing and magnitude. The QP controllers benefit moderately because their QP-optimal torques are already near the constraint boundary in the medium-force regime, leaving a learnable residual but with less complementarity to the cloning target than the passivity controller. The LQR controller benefits least because its LIPM-based ankle correction already partially overlaps with the behaviour cloning target: the LQR is producing something structurally similar to what the MLP is trained to learn, leaving less unique residual to correct. This finding is consistent with the observation of Kasaei et al. [10] that the residual benefit is largest when the base controller and the learned component address genuinely different aspects of the control problem.

The WQP–HQP equivalence identified in Chapter 4 is preserved exactly under MLP augmentation: both produce 114/200 with identical per-seed classifications across all 200 trials. This finding confirms that the architectural equivalence is a structural property of the two formulations under these conditions, not an artefact of the base controller evaluation, and that it persists regardless of what corrective layer is applied on top.

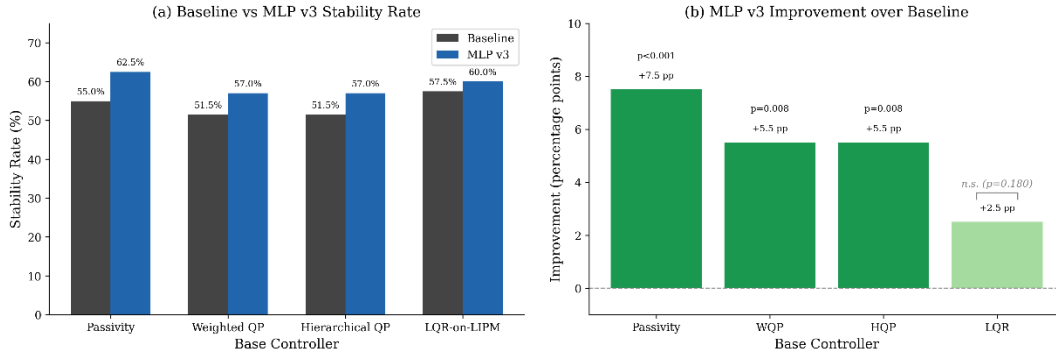


Figure 5.3 MLP v3 generalisation across all four base controllers. **Left:** baseline and MLP v3 stability rates. **Right:** improvement in percentage points per controller. Improvement correlates with the complementarity between the base controller's error structure and the pitch-rate behaviour cloning target, not with the baseline stability rate. LQR improvement is not statistically significant (n.s., $p = 0.180$).

5.4.3 Force Bin Analysis

The force-dependent improvement profile provides the most informative view of when the residual MLP adds value. For the passivity+MLP v3 controller, improvement across the four force bins is as follows. Below 75 N, the improvement is negligible because the base controller already handles these disturbances successfully. Between 75 and 125 N, improvement is +6 pp. Between 125 and 175 N, improvement reaches +13 pp — the largest gain across all bins. Above 175 N, improvement is +7 pp, but the absolute stability rate remains low (approximately 23%) because ankle torque saturation limits what any additive correction within ± 20 Nm can achieve.

The concentration of improvement in the 125–175 N range is consistent with Hypothesis H3 from Section 1.2 and with the prediction of Kasaei et al. [10] that residual learning provides greatest benefit at the boundary of the base controller's feasibility region. This is the force range in which the base controller is generating near-maximum ankle torques but has not yet fully saturated, meaning there remains a margin within which the MLP correction can influence the outcome. Below this range the base controller succeeds without assistance; above it, saturation removes the physical possibility of improvement through additive torque correction alone.

5.4.4 Cross-Controller Distillation

The cross-controller distillation experiment, in which a WQP controller is trained to reproduce the torque output of the passivity controller on matched seeds, achieves 125/200 stable trials (62.5%) compared to the WQP baseline of 103/200 (51.5%), an improvement of +11.0 pp. This result is presented separately from the residual MLP results because it addresses a fundamentally different scientific question.

Residual compensation asks: can a learned correction improve the performance of a given controller by compensating for its model-plant mismatch? Cross-controller distillation asks: can one controller be trained to approximate the torque output of a better-performing controller through supervised learning? These are different problems with different scientific implications. The distillation result confirms that the performance advantage of the passivity controller over the WQP is learnable in principle — the information needed to produce passivity-like behaviour is present in the state and is accessible to a supervised learner. However, distillation requires a target controller whose output can be used as training labels, which means it cannot improve the best-performing controller and cannot be applied when

no superior reference exists. Residual compensation does not have this limitation in principle, though as demonstrated by the v1 and v2 null results, it requires a physically grounded loss function to be effective.

5.5 Discussion

5.5.1 Why MLP v1 and v2 Failed

The null results from v1 and v2 are not incidental failures attributable to poor hyperparameter choices. They reflect a fundamental mismatch between the LIPM prediction loss and the physical objective of balance recovery. The LIPM is an unstable dynamical system — its discrete-time eigenvalue exceeds unity — and supervised learning with an unstable prediction loss produces a loss landscape in which the gradient points in the wrong direction for physically meaningful corrections. This is a theoretical result that holds regardless of network architecture, learning rate, or regularisation, and is verified numerically through the ± 50 Nm gradient analysis described in Section 5.3.1.

This finding has broader implications for residual learning in humanoid balance beyond this thesis. Any supervised loss function based on LIPM prediction will share this failure mode, because the instability is a property of the LIPM dynamics themselves rather than of a specific implementation. Researchers seeking to apply supervised residual learning to humanoid balance should avoid LIPM-based prediction losses and instead use physically grounded targets derived from known balance strategies, as demonstrated by the v3 behaviour cloning target.

5.5.2 Why MLP v3 Succeeded

The success of v3 relative to v1 and v2 is attributable to three concurrent changes, each of which addresses a specific identified limitation. The behaviour cloning target provides a non-zero, informative gradient signal derived from the ankle strategy of Stephens [17], eliminating the loss landscape problem that caused v1 and v2 to learn zero corrections. The five-frame temporal history enables disturbance detection through state trajectory observation rather than instantaneous state, addressing the push detectability problem that limits single-timestep inputs. The inclusion of training data from all 200 trials, including fallen ones, provides high-stress disturbance response data in the training set rather than restricting learning to the lower-stress conditions present in stable trials only.

The interaction between these three changes is important: no single change alone would have been sufficient. A behaviour cloning target without history would still fail to detect pushes reliably. History without a proper training target would still produce near-zero corrections under the LIPM loss. Training on all trials without the history-augmented input would not use the additional data effectively. The combination of all three changes produces the positive result observed.

5.5.3 Remaining Gap and Future Directions

Despite the statistically significant improvement, the residual MLP does not close the performance gap above 175 N. This is expected: the ± 20 Nm output bound of the MLP is insufficient to overcome ankle torque saturation, which removes the physical possibility of ankle-strategy recovery above this force threshold regardless of what additive correction is applied. As established in Section 2.4 through the capture point framework of Pratt et al. [15], recovery from disturbances above this threshold requires either stepping — which is

outside the scope of this thesis — or a fundamentally different strategy such as whole-body momentum regulation.

Three directions are identified for future work. First, reinforcement learning fine-tuning of the v3 supervised model would allow the network to explore corrections beyond the behaviour cloning target by optimising directly for the survival time reward through environment interaction, following the residual RL framework of Johannink et al. [9]. Second, a longer temporal history window or a recurrent architecture such as an LSTM would provide richer context for disturbance detection and could improve performance in the multiple-push regime identified in Section 4.3.6. Third, a stepping controller integrated with the WBC would extend the recoverable force range beyond the ankle saturation boundary, addressing the performance ceiling that no ankle-strategy controller can overcome.

5.6 Summary

This chapter investigated three supervised MLP formulations for residual learning augmentation of four WBC base controllers. Two formulations using a LIPM prediction loss produced null results, and the theoretical reason was identified and verified numerically: the LIPM's unstable discrete-time eigenvalue makes the loss gradient near-zero for physically meaningful corrections regardless of input representation. A third formulation incorporating five-frame temporal history, base controller torque as contextual input, and a pitch-rate behaviour cloning target derived from the ankle strategy of Stephens [17] produced a statistically significant improvement of 7.5 percentage points over the passivity baseline ($p < 0.001$, McNemar's exact test), generalising across all four base controllers with improvements of +5.5 pp for both QP controllers and +2.5 pp for LQR. The improvement is largest in the 125–175 N force range, consistent with the prediction that residual learning is most effective near the base controller's feasibility boundary. The null result is not a failure of the residual learning approach but a finding about loss function design: supervised residual learning for humanoid balance requires a physically grounded target that is non-zero during disturbance events, and the LIPM prediction loss does not satisfy this requirement.

6 Conclusion

6.1 Summary of Work

This thesis presented a simulation-based comparative evaluation of four whole-body control architectures for humanoid robot balance under controlled disturbance conditions, followed by a systematic investigation of supervised residual learning augmentation. The evaluation was conducted on the Unitree H1 humanoid in MuJoCo [12], covering a weighted quadratic program controller, a hierarchical quadratic program controller, a passivity-based controller, and an LQR-on-LIPM controller. Two disturbance protocols were employed: a deterministic step-push grid for baseline characterisation and a 200-seed stochastic multi-push evaluation for statistically rigorous comparison. A three-category outcome classification — stable, fell, and neither — was introduced to capture a class of lingering oscillation outcomes absent from the binary frameworks used in prior benchmarking work [7]. Three supervised MLP formulations were evaluated in progressive ablation, with the final formulation producing a statistically significant improvement in balance robustness across all four base controllers.

The work was structured in four sequential phases as described in Chapter 3, progressing from individual controller validation through comparative evaluation to residual learning augmentation. All experimental data were collected from a total of 800 base controller trials in the primary stochastic evaluation, 800 trials in the extended lower-body evaluation, and 800 trials across the four MLP v3 evaluations, producing a total of approximately 2,400 independent simulation trials. All results are fully reproducible through the seeded disturbance generation system described in Section 3.4.2.

6.2 Answers to Research Questions

The three research questions stated in Section 1.3 are answered as follows

Research Question 1 asked how the four WBC architectures compare in terms of stability rate, CoM displacement, survival time, torque effort, and computational cost under a standardised stochastic disturbance protocol. The primary finding is that the four architectures differ measurably across all five dimensions, but not always in the direction predicted by architectural complexity. The LQR-on-LIPM controller achieves the highest raw stability rate (57.5%) due to near-zero computational overhead and effective response at low disturbance forces, despite using the simplest model. The passivity-based controller achieves the best CoM quality on stable trials — 11% lower mean maximum CoM displacement than both QP controllers — and the fewest Neither outcomes (4.5% versus 7.0%), attributable to the energy-dissipating property of its damping injection layer. Both QP controllers achieve equivalent stability rates (51.5%) with the hierarchical formulation incurring 1.89 times the computational cost of the weighted formulation. All controllers converge to the same 16% stability floor above 175 N due to ankle torque saturation, confirming the physical boundary predicted by the capture point framework of Pratt et al. [15].

Research Question 2 asked whether the strict priority structure of hierarchical QP produces measurably different performance compared to weighted QP in static balance push-recovery conditions. The answer is no. Both controllers produce identical stability outcomes

on all 200 seeds, with a mean per-trial CoM difference of 0.50 mm. This equivalence arises because the 2000:1 balance-to-posture weight ratio makes the balance objective effectively dominant in all non-saturating conditions, meaning the strict priority structure of HQP is never exercised. The implementation correctness of HQP is confirmed by a divergence test showing up to 200 Nm per-joint difference under equal task weighting. The result is consistent with the theoretical prediction of Escande and Mansard [3] and the experimental confirmation of Ramuzat et al. [8] on the TALOS platform. The practical implication is that HQP's additional computational cost of 0.74 ms per step is not justified in the static balance regime, and the architectural advantage of strict priority emerges only when control tasks genuinely conflict — a condition that requires dynamic tasks such as simultaneous manipulation and balance recovery to be present.

Research Question 3 asked whether a supervised residual MLP can improve balance robustness and what training formulation properties are necessary for improvement to emerge. The answer is yes, under specific conditions. Two formulations using a LIPM prediction loss produced null results for a theoretically identified and numerically verified reason: the LIPM's unstable discrete-time eigenvalue of 1.003 produces a near-zero loss gradient for any physically meaningful correction, causing the network to learn zero corrections regardless of input. A third formulation incorporating five-frame temporal history, base controller torques as contextual input, and a pitch-rate behaviour cloning target derived from the ankle strategy of Stephens [17] produced a statistically significant improvement of 7.5 percentage points over the passivity baseline ($p < 0.001$ by McNemar's exact test). This improvement generalises across all four base controllers, is largest in the 125–175 N force range where the base controller operates near its feasibility boundary, and does not destabilise any previously stable trial. The necessary formulation properties are a physically grounded non-zero training target, temporal context for disturbance detection, and training data drawn from the full disturbance distribution including pre-fall states.

6.3 Scientific Contributions

Three scientific contributions are made by this thesis, as stated in Section 1.4.

The first is a four-architecture WBC comparative evaluation framework that produces reproducible, statistically grounded comparisons across controllers of fundamentally different types — optimisation-based, energy-based, and reduced-order — under a common stochastic disturbance protocol on the Unitree H1. The framework addresses the absence of such a comparison in the existing simulation-based benchmarking literature identified in Section 2.7.

The second is the empirical demonstration and theoretical explanation of WQP–HQP equivalence under non-conflicting task conditions. The result extends the experimental confirmation of Ramuzat et al. [8] to a different platform and a stochastic evaluation protocol, providing stronger evidence that the equivalence is a general property of the two formulations rather than a platform-specific observation. The 200/200 seed-level agreement under genuinely independent stochastic disturbances makes this one of the most statistically robust results in the thesis.

The third is the progressive ablation study of supervised residual learning, which contributes both a positive result and a theoretically grounded null result. The null result — identifying the LIPM prediction loss as fundamentally unsuitable for supervised residual learning due to the unstable eigenvalue property — is a practically useful finding for researchers applying residual learning to humanoid balance. The positive result demonstrates that supervised residual learning is feasible with the correct formulation, and the generalisation across

four architecturally distinct base controllers provides evidence that the approach is not specific to a single controller type.

6.4 Critical Discussion

Several aspects of the methodology warrant critical reflection. The evaluation is conducted entirely in simulation, and no sim-to-real validation is performed. MuJoCo's contact model, while high-fidelity, does not fully reproduce the friction variability, actuator compliance, and sensor noise present in real hardware. The extent to which the architectural rankings observed here — passivity outperforming QP controllers on stable trial quality, LQR leading on raw stability at low forces — would be preserved on the physical Unitree H1 is unknown and represents the primary uncertainty in the results. Sovukluk et al. [6] found that passivity-based WBC is more robust to modelling errors than inverse-dynamics WBC on hardware, which is consistent with the passivity controller's superior stable trial quality observed here, but a direct replication on the H1 hardware would be necessary to confirm this.

The evaluation is restricted to static balance under push-recovery conditions beginning from a single nominal starting posture. This restriction, while standard in push-recovery benchmarking [7], means the results characterise disturbance rejection from equilibrium rather than robustness across diverse initial conditions or during dynamic tasks. The architectural rankings could differ under dynamic conditions: HQP's strict priority structure may provide measurable advantage during simultaneous manipulation and balance tasks, as suggested by the theory of Escande and Mansard [3], but this was not tested here. Similarly, the LQR controller's performance advantage at low forces may not persist during walking, where the constant-height assumption of the LIPM is more severely violated.

The stochastic evaluation uses 200 seeds per controller, providing approximately 70% statistical power at $\alpha = 0.05$ for the 5–7 percentage point differences observed. This power level is appropriate for a master's-level simulation study but means that smaller effects below approximately 4 percentage points may be present without being detected. The LQR+MLP v3 improvement of 2.5 pp falls into this undetected category, and a larger evaluation with 500 or more seeds would be needed to determine whether this improvement is real or attributable to sampling variation.

The supervised residual MLP training uses a behaviour cloning target derived from a specific balance strategy — the ankle strategy of Stephens [17]. While this target is physically motivated and effective, it does not account for hip strategy responses, which become increasingly important at higher disturbance magnitudes where the ankle strategy alone is insufficient. A richer target incorporating both ankle and hip strategy components might improve performance further, particularly in the 125–175 N range.

6.5 Generalisation of Results

The findings of this thesis generalise with important conditions attached. The WQP–HQP equivalence result is expected to generalise to any static balance scenario in which balance and posture objectives do not conflict, regardless of the specific robot platform, provided that the balance weight ratio is sufficiently dominant. The equivalence is a mathematical consequence of the QP formulation established by Escande and Mansard [3] and is not specific to the Unitree H1. However, the precise weight ratio at which equivalence breaks down is platform-dependent and must be determined empirically for each new platform.

The ankle saturation finding — that all ankle-strategy controllers converge to the same failure floor above a platform-specific force threshold — generalises to any fixed-stance humanoid with limited ankle torque authority. The specific threshold of 175 N is particular to the Unitree H1's ± 40 Nm ankle limit and 47 kg mass, but the existence of such a threshold and its determination through the capture point framework of Pratt et al. [15] applies broadly.

The residual learning findings generalise with caution. The null result of LIPM-based loss functions applies to any humanoid whose LIPM has an unstable discrete-time eigenvalue, which is the case for all humanoids standing at heights producing $\omega_0 < \sqrt{2/dt}$ — satisfied by essentially all human-scale humanoids at standard simulation timesteps. The positive result of the behaviour cloning formulation is expected to generalise to other platforms, but the specific target magnitudes (8 Nm ankle, 4 Nm hip pitch) are calibrated for the Unitree H1 and would require adjustment for robots with different mass distributions and torque limits.

6.6 Future Work

Four directions for future work are identified based on the findings and limitations of this thesis.

The most important direction is hardware validation on the physical Unitree H1. The simulation results provide a controlled reference point, but architectural rankings may shift on hardware due to friction variability, actuator compliance, and sensor noise. Deploying the passivity and LQR controllers alongside the WQP on the physical platform and measuring stability rates under comparable disturbance conditions would directly test the generalisability of the simulation findings and is the natural next step from this work.

The second direction is evaluating the four controllers under conflicting task conditions, specifically simultaneous balance recovery and arm-reach tracking. As established in Section 2.2, HQP's strict priority structure provides its theoretical advantage precisely when tasks conflict. The static balance evaluation in this thesis deliberately avoids such conditions to isolate the architectural comparison, but a follow-up study with a simultaneous manipulation task would test the specific regime in which HQP is expected to outperform WQP, completing the architectural characterisation.

The third direction is reinforcement learning fine-tuning of the MLP v3 supervised model. As discussed in Section 5.5.3, the RL framework of Johannink et al. [9] would allow the network to explore corrections beyond the behaviour cloning target by optimising directly for the survival time reward through environment interaction. This approach would not suffer from the loss landscape problem that caused v1 and v2 to fail, because the reward signal is derived from simulation outcomes rather than LIPM prediction, and would likely produce larger improvements particularly in the 125–175 N regime where the behaviour cloning target is already most effective.

The fourth direction is the integration of a stepping controller to extend recovery beyond the ankle saturation boundary. As established through the capture point framework of Pratt et al. [15] and confirmed empirically in Section 4.3.3, no ankle-strategy controller can recover from disturbances above approximately 175 N for the Unitree H1. A stepping controller that transitions from ankle strategy to step strategy when the capture point exits the support polygon would address this limitation, though it requires a fundamentally different control architecture incorporating footstep planning, swing-leg trajectory generation, and contact switching — representing a substantially expanded scope that is identified as a separate research effort.

7 References

- [1] F. L. Moro and L. Sentis, “Whole-body control of humanoid robots,” in *Humanoid Robotics: A Reference*, Springer, 2019, pp. 1–22.
- [2] O. Khatib, “A unified approach for motion and force control of robot manipulators: The operational space formulation,” *IEEE J. Robot. Autom.*, vol. 3, no. 1, pp. 43–53, 1987.
- [3] A. Escande, N. Mansard, and P.-B. Wieber, “Hierarchical quadratic programming: Fast online humanoid-robot motion generation,” *Int. J. Robot. Res.*, vol. 33, no. 7, pp. 1006–1028, 2014.
- [4] A. Herzog, L. Righetti, F. Grimmering, P. Pastor, and S. Schaal, “Momentum control with hierarchical inverse dynamics on a torque-controlled humanoid,” *Auton. Robots*, vol. 40, no. 3, pp. 473–491, 2016.
- [5] B. Henze, M. A. Roa, and C. Ott, “Passivity-based whole-body balancing for torque-controlled humanoid robots in multi-contact scenarios,” *Int. J. Robot. Res.*, vol. 35, no. 12, pp. 1522–1543, 2016.
- [6] S. Sovukluk, G. Zambella, T. Egle, and C. Ott, “Experimental comparison of whole-body control formulations for humanoid robots in task acceleration and task force spaces,” *arXiv:2507.18502*, 2025.
- [7] J. A. Castano, A. Chemori, J. Andrade-Chavez, and J. Maldonado, “Benchmarking dynamic balancing controllers for humanoid robots,” *Sensors*, vol. 22, no. 11, p. 4226, 2022.
- [8] N. Ramuzat, G. Buondonno, S. Boria, and O. Stasse, “Benchmarking whole-body controllers on the TALOS humanoid robot,” *Front. Robot. AI*, vol. 9, p. 807006, 2022.
- [9] T. Johannink, S. Bahl, A. Nair, J. Luo, A. Kumar, M. Loskyll, J. A. Ojea, E. Solowjow, and S. Levine, “Residual reinforcement learning for robot control,” in *Proc. IEEE Int. Conf. Robot. Autom.*, Montreal, Canada, 2019, pp. 6023–6029.
- [10] M. Kasaei, M. Sotirchos, N. Lau, and A. Pereira, “Learning hybrid locomotion skills: Learn to exploit residual actions and modulate model-based gait control,” *Front. Robot. AI*, vol. 10, p. 1106466, 2023.
- [11] J. Hwangbo, J. Lee, A. Dosovitskiy, D. Bellicoso, V. Tsounis, V. Koltun, and M. Hutter, “Learning agile and dynamic motor skills for legged robots,” *Sci. Robot.*, vol. 4, no. 26, p. eaau5872, 2019.
- [12] E. Todorov, T. Erez, and Y. Tassa, “MuJoCo: A physics engine for model-based control,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, Vilamoura, Portugal, 2012, pp. 5026–5033.
- [13] P. M. Wensing, M. Posa, Y. Hu, A. Escande, N. Mansard, and A. Del Prete, “Optimization-based control for dynamic legged robots,” *IEEE Trans. Robot.*, vol. 40, no. 1, pp. 43–63, 2024.
- [14] Y. Lee, S. Kim, J. Park, N. Tsagarakis, and J. Lee, “A whole-body control framework based on the operational space formulation under inequality constraints via task-oriented optimization,” *IEEE Access*, vol. 9, pp. 39813–39826, 2021.
- [15] J. Pratt, J. Carff, S. Drakunov, and A. Goswami, “Capture point: A step toward humanoid push recovery,” in *Proc. IEEE-RAS Int. Conf. Humanoid Robots*, Genova, Italy, 2006, pp. 200–207.
- [16] A. Elhasairi and A. Pechev, “Humanoid robot balance control using the spherical inverted pendulum model,” *Front. Robot. AI*, vol. 2, p. 21, 2015.
- [17] B. Stephens, “Integral control of humanoid balance,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, San Diego, CA, 2007, pp. 4020–4027.

- [18] M.-J. Kim, D. Lim, G. Park, K. Lee, and J. Park, “A model predictive capture point control framework for robust humanoid balancing via ankle, hip, and stepping strategies,” arXiv:2307.13243, 2023.
- [19] Y. Cho and J. H. Park, “Whole-body motion generation for balancing of biped robot,” *Appl. Sci.*, vol. 15, no. 1, p. 214, 2025.
- [20] S. Heng, P. Tian, and T. Hamel, “A robust disturbance rejection whole-body control framework for bipedal robots using a momentum-based observer,” *Biomimetics*, vol. 10, no. 3, p. 145, 2025.
- [21] S. Zhang, F. Ni, X. Li, X. Li, Y. Ji, and X. Shao, “Robustness and push recovery of humanoid robots using predictive and whole-body control,” in *Proc. IEEE Int. Conf. Unmanned Syst.*, Nanjing, China, 2024, pp. 871–877.
- [22] Y. Yan, E. V. Mascaró, T. Egle, and D. Lee, “I-CTRL: Imitation to control humanoid robots through bounded residual reinforcement learning,” *IEEE Robot. Autom. Mag.*, vol. 32, no. 1, pp. 59–67, 2025.
- [23] H. Wang and W. Huang, “Gait generation and motion implementation of humanoid robots based on hierarchical whole-body control,” *Electronics*, vol. 14, no. 23, p. 4714, 2025.
- [24] J. Ahn, “Time-efficient hierarchical whole-body control system framework for humanoid robot,” Ph.D. dissertation, Seoul National Univ., Seoul, South Korea, 2025.
- [25] Y. Liu, H. Mou, H. Jiang, Q. Li, and J. Zhang, “An improved hierarchical optimization framework for walking control of underactuated humanoid robots using MPC and whole-body planner and controller,” *Mathematics*, vol. 13, no. 1, p. 75, 2025.
- [26] K. Zakka et al., “MuJoCo Playground: An interactive learning environment for robot skill development,” arXiv:2502.08844, 2025.
- [27] C. Sferrazza, D.-M. Huang, X. Lin, Y. Lee, and P. Abbeel, “HumanoidBench: Simulated humanoid benchmark for whole-body locomotion and manipulation,” arXiv:2403.10506, 2024.